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PERSONAL REPORT

**THEME: IMPLEMENTATION OF A MOBILE APPLICATION FOR
PLANT DIAGNOSIS**

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CLASS: SE3A

ACADEMIC YEAR: 2024-2025



IMPLEMENTATION OF A MOBILE APPLICATION FOR PLANT DIAGNOSIS



DEDICATION

THIS BOOK IS DEDICATED TO THE
OBIANG'S FAMILY FOR THEIR LOVE AND
SUPPORT



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LIST OF ABBREVIATIONS

- **2TUP:** Two Track Unified Process
- **AICS:** African Institute of Computer Sciences
- **API:** Application Programming Interface
- **CSS:** Cascading Style Sheet
- **DBMS:** Database Management System
- **GUI:** Graphical User Interface
- **HTML:** Hypertext Markup Language
- **HTTP:** Hypertext Transfer Protocol
- **JSON:** JavaScript Object Notation
- **OS:** Operating System
- **UML:** Unified Modelling Language



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ABSTRACT

In Cameroon, many men and women turn to farming not out of choice but as a last resort. They dedicate their time, energy, and scarce resources to small plots of land, hoping to harvest enough to feed their families or earn a modest income. Yet these efforts are often ruined by plant diseases that remain unnoticed or untreated until it is too late. For such farmers, losing a harvest means losing their last chance at stability. This report presents the theme: **Analysis and implementation of a plant health diagnosis application**. The aim is to design a simple but effective tool that allows farmers to quickly identify plant diseases, understand treatments in clear language, and contact agronomists at a reduced cost when needed. The application combines **Artificial Intelligence**, which analyzes plant images to suggest possible diseases, with an **expert consultation module** that connects farmers with professionals. Unlike platforms full of technical jargon, this solution is adapted to farmers with little or no formal agricultural training.

Through this project, technology becomes more than a tool — it becomes a partner: helping to protect crops, reduce losses, and restore hope. More than a prototype, this work represents a step toward modernizing agriculture while giving back dignity and confidence to those who depend on the land for survival.

Keywords:

- Plant diagnosis
- Artificial Intelligence
- Expert system
- Agriculture
- Mobile application

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RESUME

Au Cameroun, beaucoup d'hommes et de femmes se tournent vers l'agriculture non pas par choix, mais comme dernier recours. Ils consacrent leur temps, leur énergie et leurs maigres ressources à de petites parcelles de terre, espérant récolter assez pour nourrir leurs familles ou obtenir un revenu modeste. Pourtant, ces efforts sont souvent anéantis par des maladies des plantes, détectées trop tard ou mal soignées. Pour ces agriculteurs, perdre une récolte signifie perdre leur dernière chance de stabilité. C'est dans ce contexte que s'inscrit le présent rapport, sur le thème : *« Analyse et implémentation d'une application de diagnostic de santé des plantes. »* L'objectif est de concevoir un outil simple mais efficace, permettant aux agriculteurs d'identifier rapidement les maladies, de comprendre les traitements dans un langage clair et d'accéder à des agronomes à moindre coût en cas de besoin. L'application associe l'Intelligence Artificielle, qui analyse les images des plantes pour proposer un diagnostic, à un module de consultation d'experts. Contrairement aux plateformes complexes remplies de jargon technique, cette solution est pensée pour des agriculteurs sans formation agricole formelle. Ainsi, la technologie devient plus qu'un outil : elle devient un partenaire, protégeant les cultures, réduisant les pertes et redonnant espoir. Plus qu'un simple prototype, ce projet représente un pas vers la modernisation de l'agriculture, tout en rendant dignité et confiance à ceux qui dépendent de la terre pour survivre.

Mots-clés:

- Diagnostic des plantes
- Intelligence Artificielle
- Système expert
- Agriculture
- Application mobile



GENERAL INTRODUCTION

Agriculture is one of the most important sectors of Cameroon's economy, employing the majority of the population and contributing significantly to national growth. However, farmers continue to face many challenges, among which plant diseases remain a critical problem. These diseases reduce productivity, increase costs, and limit food security. In a context where access to rapid and affordable diagnosis is often lacking, the use of digital technologies becomes a promising solution. The **African Institute of Computer Sciences**, in its mission of training highly skilled engineers, encourages students to bridge theoretical knowledge with real-life problems through academic internships. This initiative aims at preparing students for professional challenges while fostering innovation that impacts society positively.

It is within this framework that I carried out my academic internship at the startup **REALIZE**, a young and dynamic enterprise specializing in the development of digital solutions for individuals and organizations. During this internship, I worked on the theme: **Analysis and implementation of a plant health diagnosis application**. This application aims to provide farmers with quick and reliable plant disease diagnosis and the possibility of contacting agricultural experts at a reduced cost. To achieve these objectives, this report is structured into several phases:

- **Part I:** The Existing System
- **Part II:** The Specification Phase
- **Part III:** The Analysis Phase
- **Part IV:** The Conception Phase
- **Part V:** The Realization Phase
- **Part VI:** Test of Functionalities
- **Part VII:** User Guide



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PART I: EXISTING SYSTEM

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Preamble

The existing system is a document that provides a view of how the current system works, how it carries out its different activities, it equally provides a deep understanding of the system associated to the various limitations and the problems that result from it and we propose the solution.

Content

INTRODUCTION

- I. PRESENTATION OF THE PROJECT THEME
- II. DESCRIPTION OF THE EXISTING SYSTEM
- III. CRITICISM OF THE EXISTING SYSTEM
- IV. PROBLEMATICS
- V. PROPOSED SOLUTION

CONCUSION

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IMPLEMENTATION OF A MOBILE APPLICATION FOR PLANT DIAGNOSIS



INTRODUCTION

This section explores how farmers in Cameroon and similar agricultural contexts currently diagnose and manage plant diseases. It describes the informal methods based on personal experience or peer consultation, as well as the use of limited mobile tools such as Plantix. By examining these practices, we highlight their shortcomings in terms of accuracy, accessibility, and sustainability. Identifying these gaps is essential, as they form the foundation of the problem statement motivating this research. Furthermore, this analysis provides a framework for justifying the relevance of our proposed digital solution. The ultimate goal is to establish why a new system such as HARVEST is necessary and how it responds to these challenges.



IMPLEMENTATION OF A MOBILE APPLICATION FOR PLANT DIAGNOSIS



I. PRESENTATION OF THE PROJECT THEME

The project entitled **Analysis and Implementation of a Mobile Application for Plant Health Diagnosis** focuses on developing **HARVEST**, a mobile tool designed to assist farmers in identifying plant diseases quickly and accurately. Unlike conventional methods or generic applications, HARVEST is designed specifically for the Cameroonian agricultural context, with an emphasis on simplicity and adaptability to regional crops and diseases.

II. DESCRIPTION OF THE EXISTING SYSTEM

Currently, farmers in Cameroon use a variety of methods to assess plant health:

- **Experience-based diagnosis:** Agronomists or experienced farmers rely on their personal knowledge to identify diseases.
- **Informal consultations:** Farmers without prior expertise often seek advice from neighbors or community members.
- **Online research:** Some turn to the internet or social media platforms to look for similar cases and possible solutions.
- **Mobile applications:** Certain farmers use apps like **Plantix**, which allows users to take a photo of the plant and receive a diagnosis generated by its database. Plantix operates by comparing the uploaded image to its dataset of plant diseases. It provides a diagnosis and recommended treatments based on predefined information, without further interaction with the user.



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III. CRITICISM OF THE EXISTING SYSTEM

Table 1: Weakness and Impact on users of the existing system

LIMITATIONS	CONSEQUENCES	PROPOSED SOLUTION
Experience-based diagnosis required	Farmers without prior knowledge are unable to identify diseases accurately, leading to delays in treatment.	Provide an AI-powered application that can guide diagnosis without requiring prior expertise.
Risk of misdiagnosis	Wrong treatments applied, leading to crop loss and reduced yields.	Integrate an AI system that asks context-related questions to improve accuracy.
Limited access to agronomists / high consultation costs	Farmers in rural areas cannot afford or reach professional experts easily.	Ensure offline access and a user-friendly tool that can be used directly in the field.
No record of previous diagnoses	Farmers cannot track recurring issues or monitor the effectiveness of past treatments.	Add a history feature to save and review previous diagnoses.
Plantix hardcoded responses	Limited adaptability; results not tailored to each case.	Design HARVEST to be interactive and adaptable to context.
Complex language in Plantix	Farmers may not understand results and treatments.	Use simplified and locally adapted language for better comprehension
Not adapted to local crop diseases	Ineffective for diagnosing region-specific problems.	Build a database focused on Cameroonian crops and diseases.

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IV. PROBLEMATIC

How can farmers in Cameroon obtain a **fast, accurate, and affordable diagnosis** of plant diseases while also gaining access to expert advice when necessary?

V. PROPOSED SOLUTIONS

The proposed solution is the **HARVEST** mobile application. It integrates an AI-powered plant health diagnosis system capable of:

- Capturing plant images for analysis.
- Asking context-related questions to improve diagnostic accuracy.
- Delivering results in simple, locally understandable language.
- Operating offline to accommodate farmers in low-connectivity areas.

But also, it provides access to competent and experienced experts at a reduced cost.

The goal is to offer an accessible, user-friendly tool that gives farmers a reliable starting point for diagnosing plant diseases and making informed decisions to protect their crops.



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CONCLUSION

The current methods of diagnosing plant diseases are limited and insufficient to meet the needs of farmers. This project aims to overcome these barriers by providing a digital solution that combines Artificial Intelligence with expert consultation, ensuring fast, reliable, and accessible plant health management.

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PART II: SPECIFICATION BOOK

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IMPLEMENTATION OF A MOBILE APPLICATION FOR PLANT DIAGNOSIS



Preamble

The specification book defines the needs and objectives of the project while establishing the resources and constraints necessary for its realization. It ensures a clear understanding between the users' expectations and the technical solution that must be provided.

Content

INTRODUCTION

I. CONTEXT AND JUSTIFICATION

II. OBJECTIVES OF THE PROJECT

III. EXPRESSION OF NEEDS

IV. PROJECT PLANNING

V. ESTIMATED COST OF THE PROJECT

VI. PROJECT CONSTRAINTS

VII. DELIVERABLES CONCLUSION

CONCLUSION

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IMPLEMENTATION OF A MOBILE APPLICATION FOR PLANT DIAGNOSIS



INTRODUCTION

The specification book of our reports helps us provide details about our theme, to improve our understanding of it and increase the likelihood of it succeeding. To delimitate the scope of our project, we will specify the context of our theme. From the context, we will list the problems we have identified in our context and that we have decided to address throughout the project. After presenting our solution, we will talk about the objective we have set for the project. Also, we will explore the needs to which our system will respond both at the functional and non-functional level. We will then look at the estimated financial requirements for our project, and establish a plan we will follow to complete our project on time. From here we will discuss what is expected of us by the end of the project under the project deliverables.



I. CONTEXT AND JUSTIFICATION

1. CONTEXT

In Cameroon, farming isn't just a job it's the backbone of daily life. It puts food on the table for most families and keeps rural communities alive. But for the farmers doing this vital work, every season is a gamble. One undetected pest, one sudden disease, or a stretch of bad weather can wipe out months of hard work in days. Right now, most farmers have to rely on gut instinct, word-of-mouth from neighbors, or a rare, expensive trip to see an expert methods that are slow, hit-or-miss, and often too late. The frustrating part? The solution is probably already in their pocket. With mobile phones now common even in remote villages, we have a real chance to give farmers instant, reliable help right when they need it most.

2. JUSTIFICATION

Here's the harsh reality for farmers in Cameroon's remote communities: when a crop starts looking sick, the clock starts ticking and it's rarely in their favor. Getting a proper diagnosis isn't as simple as making a call or sending a photo. Instead, it means waiting days or even weeks for an agricultural extension officer to show up if they show up at all. And even then, the diagnosis often boils down to a quick visual guess, with no real system behind it. That guesswork is dangerous. Misidentifying a disease doesn't just waste time; it can actually make things worse. Farmers might apply the wrong treatment spending hard-earned money on something that doesn't work, harming their soil, and watching the disease spread to neighboring plots while they wait. None of this is the farmer's fault. They're doing everything they can with what they have. The real problem is a broken system: one that's too slow, too centralized, and too reliant on a handful of experts covering vast territories. But what if that system could fit inside a pocket? Smartphones are becoming common even in rural villages, and artificial intelligence has reached a point where it can "see" patterns in a single leaf photo faster and more accurately than the human eye. That's the gap we're filling. Our mobile app isn't just a convenience it's a lifeline. By putting an AI-powered diagnostic tool directly into farmers' hands, we bypass the long waits, the long distances, and the guesswork. Suddenly, an expert is available 24/7, right where the crops are growing. No roads, no appointments, no delays just answers, when they matter most.



II. OBJECTIVE OF THE PROJECT

A. GENERAL OBJECTIVE

To design, develop, and deploy a user-friendly and accessible mobile application that provides instant diagnosis and treatment recommendations for plant diseases, empowering Cameroonian farmers and improving agricultural productivity.

B. SPECIFIC OBJECTIVES

- To create a mobile-based application that allows users to capture images of diseased plants and receive a real-time diagnosis.
- To integrate a treatment recommendation system that provides clear, actionable advice on how to manage and cure identified diseases.
- To ensure the application is scalable and maintainable to allow for the future addition of new crops and diseases.
- To design an intuitive and simple user interface that is usable by farmers with a wide range of digital literacy.
- Offer an integrated platform where farmers can consult agronomists at reduced cost.



III. EXPRESSION OF NEEDS

1. FUNCTIONAL NEEDS

These are the core functions the system must provide to users:

For Farmer:

- ❖ Image Capture: The ability to take and upload a photo of a plant leaf.
- ❖ Instant Diagnosis: The application must analyze the image and provide an immediate diagnosis.
- ❖ Treatment Recommendations: Display detailed information about the disease and its corresponding treatment methods (both chemical and organic).
- ❖ Expert Consultation: The ability to submit a case for review by a certified agronomist for a second opinion or complex diagnosis.
- ❖ Offline Functionality: The core diagnostic feature should work without an internet connection.

For Expert:

- ❖ Case Review: Access to a dedicated dashboard to view and manage farmer-submitted cases.
- ❖ Expert Feedback: The ability to provide detailed written feedback and customized treatment recommendations to a farmer.
- ❖ Knowledge Base Contribution: The ability to add new plant diseases, treatment methods, or update existing information.

For Administrators:

- ❖ Admin Dashboard: Access to a web-based dashboard for managing the application.
- ❖ User Management: The ability to manage and verify user accounts, expert profile.
- ❖ Analytics: Monitor user engagement, most common diagnoses, and other key metrics.
- ❖ Content Management: The ability to add, edit, or remove plant diseases and treatment information.



2. NON-FUNCTIONAL NEEDS

These define the performance and quality aspects of the system:

Performance:

- ❖ Fast Response Time: Diagnosis must be delivered in under 10 seconds.
- ❖ Smooth Performance: The app should run efficiently on low-end smartphones with minimal lag.
- ❖ Reliability & Accuracy:
- ❖ High Accuracy: The ML model must achieve at least 95% accuracy in disease identification.
- ❖ Data Security: All user data, including images, must be securely handled and anonymized.

Usability:

- ❖ Simple Interface: The UI must be intuitive and require minimal training for a rural user base.
- ❖ Mobile-Friendly: The design must be optimized for a variety of screen sizes and resolutions.

Scalability & Maintainability:

- ❖ Modular Architecture: The system should be designed to easily add new crops and diseases as the project expands.
- ❖ Clean Codebase: The code must be well-documented and easy to maintain for future development.

IV. PROJECT PLANNING

A. CHRONOGRAM OF ACTIVITIES

Table 2: Chronogram of activities

No	Activity	Duration	Start Date	End Date
1	Requirements gathering & research	2 weeks	01/01/2026	16/01/2026
2	System analysis & UML modelling	2 weeks	14/01/2026	30/01/2026
3	Application design	2 weeks	31/01/2026	13/02/2026
4	AI image analysis & questioning system	3 weeks	14/02/2026	03/03/2026
5	Language & expert module integration	1 week	04/03/2026	10/03/2026
6	Testing & debugging	1 week	11/03/2026	17/03/2026
7	Deployment & user guide preparation	1 week	18/03/2026	24/03/2026
8	Final report writing & submission	1 week	25/03/2025	30/03/2026



V. ESTIMATED COST OF PROJECT

A. Software resources

Table 3: Software Resources

Resource	Type	Estimated Cost	Notes
Node.js + Express	Open-source	0 FCFA	Backend development
React native js	Open-source	0 FCFA	Frontend development
MongoDB	Free tier	0 FCFA	Suitable for MVP
GitHub	Free	0 FCFA	Version control
Vercel/ Render (Free tier)	Free	0 FCFA	Hosting/deployment for MVP
AI Tools (Gemini AI API)	Freemium	30,000 FCFA	For smart search (approx. testing)
Total 1		30,000 FCFA	

B. Hardware resources

Table 4: Hardware Resources

Resource	Quantity	Estimated Cost (FCFA)	Notes
Developer laptop	1	500,000 FCFA	Minimum specs: i5, 16GB RAM, 512 SSD
Internet connection	3 months	~30,000 FCFA	10,000/month
Smartphone (testing)	1(optional)	Already owned or ~110,000 FCFA	For responsive design tests
Total 2		~30,000-700,000 FCFA	



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C. Human resources

Table 5: Human Resources

Role	Duration	Estimated (FCFA)	Cost	Notes
Full-stack developer (you)	3 months internship	50,000FCFA		
Supervisor	5 hours support	0 FCFA		Only if external expert consulted
Designer	1–2 days UI help	0 FCFA		Optional, if not done by student
Total 3		~0-50,000 FCFA		

D. Global resources

Table 6: Global resources

Category	Estimated Cost range
Software resources	0-15000
Hardware resources	30,000-400,00
Human resources	0-25,000

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VI. CONSTRAINTS

The project will be subject to several critical constraints that must be carefully managed. The first is Data Availability. Sourcing a high-quality, comprehensive, and well-labeled dataset of images is the single most challenging aspect of this project. Plant diseases can manifest differently under various environmental conditions, and we need to ensure our model is trained on a diverse set of examples to avoid bias and ensure high accuracy in the field. A limited or skewed dataset could severely compromise the reliability of our application. We plan to mitigate this by engaging local partners and dedicating sufficient time to this phase.

Another significant constraint is Internet Connectivity. While smartphone adoption is high, internet access in many rural farming communities is either nonexistent, unreliable, or very slow. This poses a direct threat to the application's functionality. We will address this by designing the application with a strong offline-first approach. The core diagnostic model will be packaged within the app itself, allowing it to function without an internet connection for real-time diagnosis. Internet will only be required for initial download, updates, and possibly for transmitting aggregated data for future model improvements.

Finally, User Adoption represents a social and cultural constraint. Introducing a new technology to a population accustomed to traditional farming methods requires more than just a functional application; it requires trust-building and education. Farmers may be skeptical of a machine-based diagnosis and may be resistant to change. To overcome this, we plan to implement a robust community engagement strategy, conducting workshops and demonstrations to show the application's benefits and usability. Additionally, the app's success is tied to Technological Limitations; it must run smoothly on a wide range of mobile devices, including low-end models with limited processing power and battery life. The ML model must be optimized to be lightweight and efficient to ensure a good user experience for all.



VII. DELIVRABLES

In project management, any component materializing the result of a realization service is called a deliverable. In the case of our project, the deliverables are: A report composed of the following document

- The application.
- The user guide.
- The PowerPoint.



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CONCLUSION

This detailed project specification document provides a comprehensive roadmap for developing a transformative technological solution for Cameroonian farmers. By deeply understanding the context, setting clear objectives, and meticulously planning for potential constraints, we have created a robust framework for success. Our team is committed to delivering a high-impact mobile application that not only addresses a critical problem but also serves as a powerful tool for empowering agricultural communities, ultimately contributing to a more prosperous and sustainable future..

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PART III: ANALYSIS PHASE



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Preamble

The main objective of the analysis phase is to capture the user's need, the delimitation of the field of study and to have a clear understanding of the system in study. To achieve this, we will use UML (Unified Modelling Language) with the 2TUP (2 Track Unified Process) as method applied to UML to analyze the system. We will start by doing a comparison between UML and MERISE, of various unified processes and lastly, we will present the modelling of the solution we propose (Our software).

Content

INTRODUCTION

I. METHODOLOGY

1. COMPARATIVE STUDY OF UML AND MERISE.
2. COMPARATIVE STUDY OF UNIFIED PROCESSES.

II. CHOICE OF THE ANALYSIS METHOD

CONCLUSION

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INTRODUCTION

System development can be thought of as having two major components: System analysis and system design which both help in understanding the details of the existing system or the system to be designed. The analysis and design of information systems has most of the time vocation to allow the creation of databases, which must represent as closely as possible the reality of the field studied thus requiring the use of a design method. This is why our choice will be directed on the UML method as it offers much to developers seeking a user-centered approach and / or a wide scope in design. This part of the report consists of the comparative study of UML and MERISE, unified processes and finally the various diagrams that meet the functional need requirements.



I. METHODOLOGY

1. COMPARATIVE STUDY OF UML AND MERISE

a. MERISE

MERISE stands for “Méthode d’Etude et de Réalisation Informatique pour des Systèmes d’Entreprise”. Although it is prescriptive to some extent, MERISE permits the participation of end users and senior management as well as data processing professionals in its decision cycle. MERISE is a method for designing, developing and carrying out IT projects. The goal of this method is to achieve the design of an information system. The MERISE method is based on the separation of data and processing to be carried out in several conceptual and physical models. The essentials of the approach lie in its three cycles: the decision cycle, the life cycle and the abstraction cycle, which cover data and process elements equally. The separation of data and processing ensures longevity in model. Indeed, the arrangement of data does not have to be often overhauled, while treatments are more frequently.

b. UML

UML (Unified Modelling Language) is a standard notation for the modelling of real world objects as a first step in developing an object-oriented design methodology. Its notation is derived from three object-oriented design and analysis methodologies: Grady Booch's methodology for describing a set of objects and their relationships, James Rumbaugh's Object-Modelling Technique (OMT), Ivar Jacobson's approach which includes a use case methodology. Other ideas also contributed to UML, which was the result of a work effort by Booch, Rumbaugh, Jacobson, and others to combine their ideas, working under the sponsorship of Rational Software. UML captures information about the static and dynamic view of a system. UML 2.5 comprises of 14 diagrams which represent the different views of a system. The 14 diagrams can be subdivided into two, Static or structural and Dynamic diagrams. These diagrams include;



II. STATIC OR STRUCTURAL DIAGRAMS

- Class diagram;
- Object diagram;
- Component diagram;
- Deployment diagram;
- Composite Structure diagram;
- Package diagram;
- Profile Diagram;

III. BEHAVIOURAL OR DYNAMIC DIAGRAMS

- Use case diagrams;
- Activity diagram;
- State machine diagram;
- Sequence diagram;
- Communication diagram;
- Interaction diagram;
- Timing Diagram;

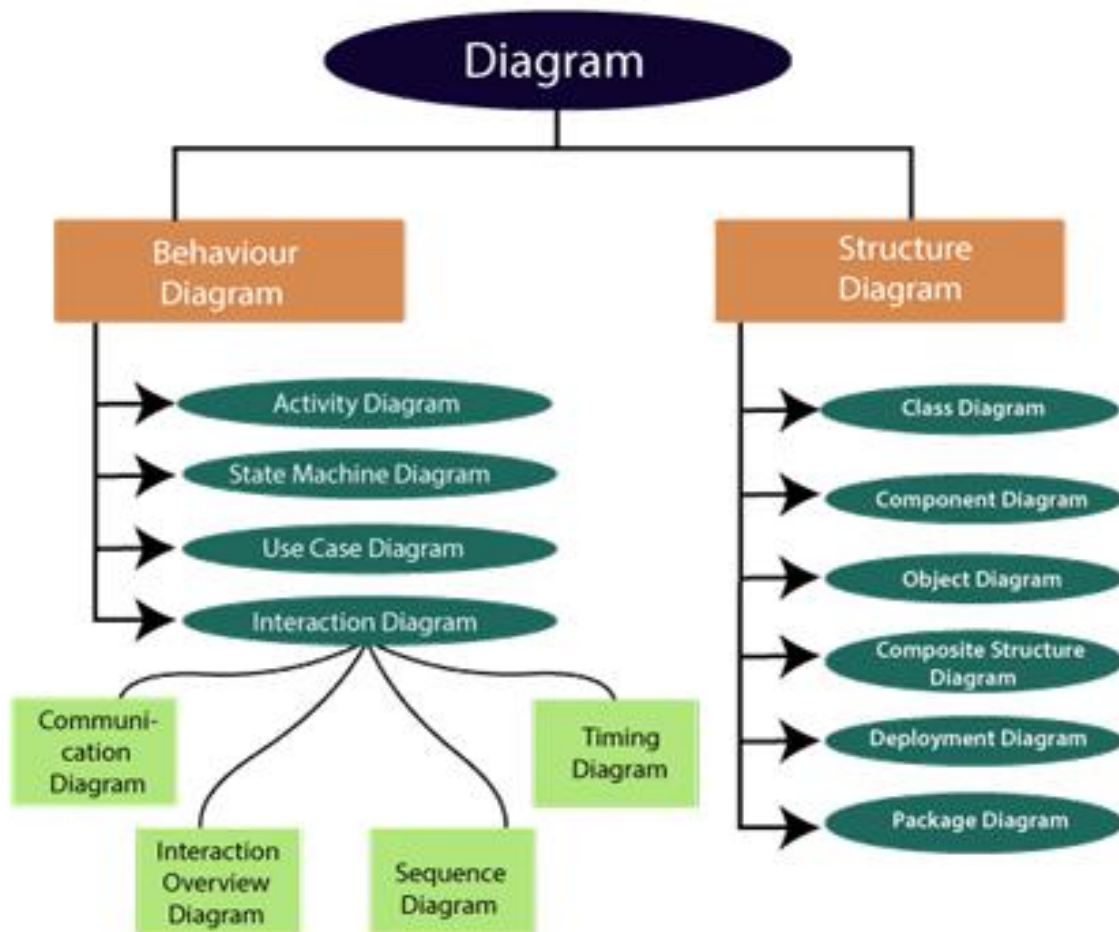


Figure 1: Uml Diagram types

Table 7: Difference between Uml and Merise

MERISE	UML
It stands for Méthode d'Étude et de Réalisation Informatique pour les Systèmes d'Entreprises	Unified Modeling Language
MERISE is a systemic method of analysis and design of information systems. That is, it uses a systems approach.	UML is however not a method but rather an object modeling language to which it is necessary to associate an approach to make it a method. This is the case with the 2TUP method; RUP and XP.
MERISE proposes to consider the real system from two points of view: - A static view (data) - A dynamic view (treatments). That is, with the MERISE method, we have a separate study of the data and the treatments.	UML offers a different approach from that of MERISE in that it combines data and processing. Because with UML, centralizing the data of a type and the associated processing makes it possible to limit the maintenance points in the code and facilitates access to information in the event of software development. In addition, UML describes the dynamics of the information system as a set of operations attached to the objects of the system.
Rational	Object



2. COMPARATIVE STUDY OF UNIFIED PROCESSES

a. A Unified Process

The UP is an abbreviation of Unified Process. It is an iterative and incremental software development methodology. The Unified Process is an iterative, architecture centric software development process driven by use cases and geared towards reducing risk. It is a process pattern that can be adapted to a wide class of software systems, to different areas of application, to different types of businesses, to different skill levels and to different sizes of the business. Different data. It qualifies a process or a procedure that performs a group of operations repeatedly until a well-defined condition is met.

b. The Two Track Unified Process (2 TUP)

2TUP is a unified process which is built on UML and has as objective to bring solution to constraints of functional and technical changes imposed on information systems by strengthening controls on development capacities. It proposes a Y-shaped development life cycle that separates the functional aspect from the technical aspects, and the merging of these two forms the implementation aspect. 2TUP distinguishes therefore two branches: the functional and technical branches, the combination of the result of these two branches forms the third: the realization branch – where we realize our system. The diagram below illustrates the branches of 2TUP.

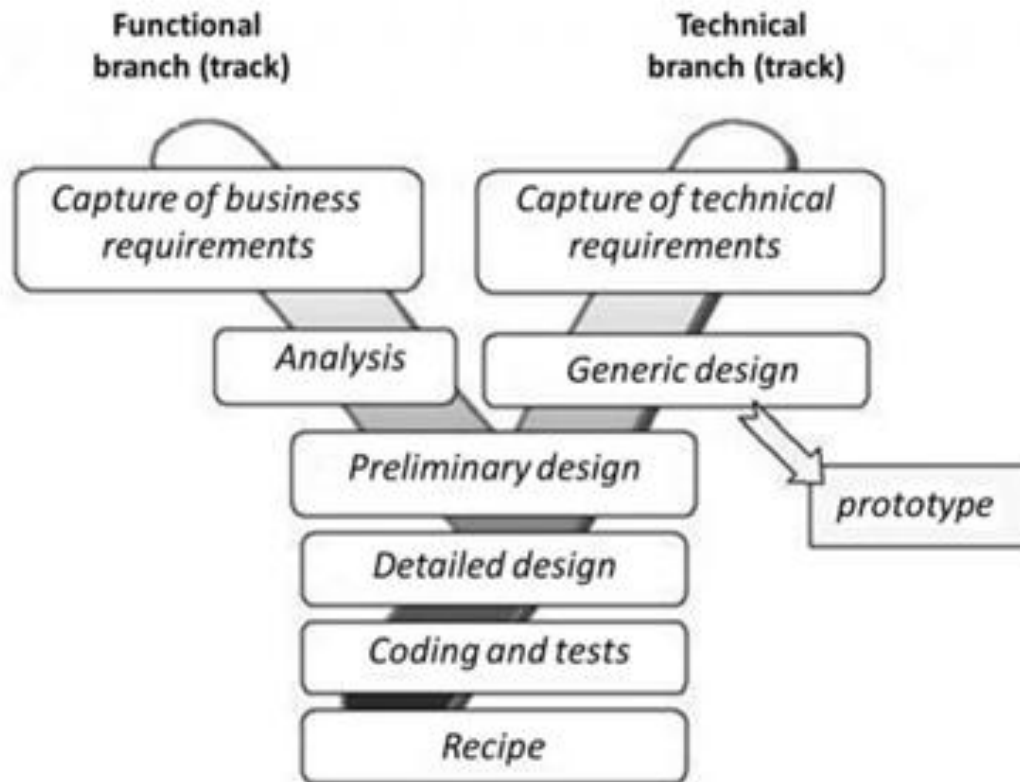


Figure 2 :2TUP diagram

a. The left branch (functional branch)

It captures the functional needs of a system. This ensures the production of software that meets the needs/requirements of the user. The analysis here consists of studying precisely the functional specification in order to obtain an idea of what the system is going to realize, and its result does not depend on any technology.

b. The right branch (Technical branch)

The technical branch enumerates the technical needs and proposes a generic design validated by a prototype. The technical needs include constraints and choices related to the conception of the system, the tool. The middle branch (Realization or Implementation branch)



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In this branch, we study the preliminary conception, detailed conception, and documentation of the system. The realization branch supports the following: Preliminary conception: This is the most sensitive step of 2TUP as it is the confluence of the functional and technical branch. It is completed when the deployment model, the operating model, the logical model, interphases and the software configuration model are defined. We have the following diagrams:

- ❖ Component Diagram;
- ❖ Deployment Diagram;
- ❖ Package Diagram;
- ❖ Composite Structure Diagram;

Detailed conception: This is the detailed design of each feature of the system. We have the following diagrams:

- ❖ Class;
- ❖ Sequence;
- ❖ Object;
- ❖ Timing diagram;

Coding and testing: This is the phase where we program the designed features and test the coded features.

The recipe: Also known as the deliverables is the validation phase of the functions of the developed system.



3. JUSTIFICATION OF THE ANALYSIS METHOD

The reason why we chose UML modelling language and the software development process 2TUP instead of many others that exist, include: UML is the current standard for programming in an object-oriented language. For this reason, it is widely understood and well known making it easy for a new programmer to join the project and be productive from the very first day. UML diagrams allow teams to virtualize how a project is or will be working, and they can be used in any field, not just software engineering. The diagrams will allow teams to virtualize together how a system, or a process will work or did work. It can provide new ideas for how teams have to collaborate to achieve the goal of the workflow process. 2TUP is centered on the creation and maintenance of a model, rather than the production of mountain documents. 2TUP is user oriented as it permits the development of software that responds to the needs of the users through the study of the user needs. 2TUP is iterative and incremental, hence it enables the project team to produce refined amelioration if necessary and easily integrate it in the already existing system. 2TUP by permitting the project team identify and test the key functionalities of the system limits the risk related to building the system.

III. MODELLING OF THE PROPOSED SOLUTION

A. Capture of Functional Needs

The first step of the left (functional) branch of Two Track Unified Process (2TUP) is the capture of the functional needs. At this step, we capture the intended behavior of the system that maybe express as services, tasks or functions the system is required to perform.

1. USE CASE DIAGRAM

a) Definition

Use-case diagrams describe the high-level functions and scope of a system. These diagrams also identify the interactions between the system and its actors. The main objectives of the use case diagram are:

- Provide a high-level view of the system.
- Identify the functions of the system.

Use case diagrams are completed with a textual description of each use case that is intended to define the use case in greater details.

b) Formalism of a use case diagram

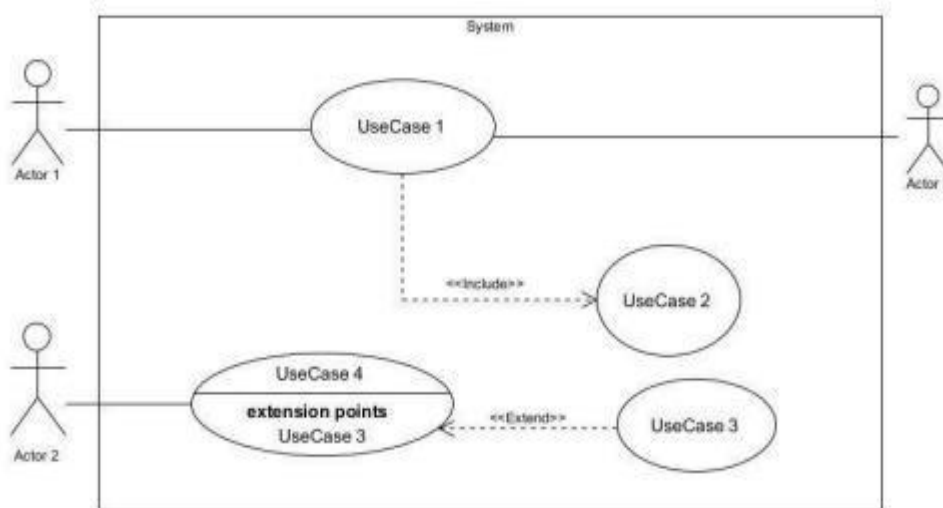


Figure 3: Use case Diagram formalism



c) System Actors

In order to model the plant health diagnosis application, it is essential to identify all the actors that will interact with the system. These actors represent both the end-users and external systems necessary for the proper functioning of the application.

1. Farmer

- **Role:** Main user of the application.
- **Responsibilities:**
 - ✓ Create an account and log in.
 - ✓ Capture or upload images of plants.
 - ✓ Receive diagnosis results.
 - ✓ Request consultations with agronomists.
 - ✓ View history of diagnoses and treatments.

2. Expert

- **Role:** Professional who provides agricultural advice.
- **Responsibilities:**
 - ✓ Respond to consultation requests.
 - ✓ Provide treatment recommendations.
 - ✓ Update personal availability.

3. Administrator

- **Role:** Ensures the smooth functioning of the platform.
- **Responsibilities:**
 - ✓ Manage farmer and expert accounts.
 - ✓ Monitor transactions and consultations.
 - ✓ Ensure system security and performance.



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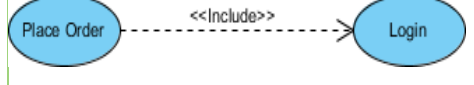
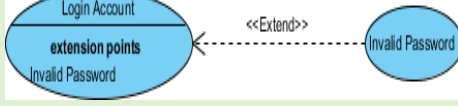
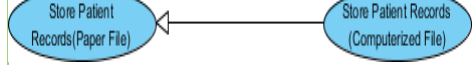


4. Payment API

- **Role:** External financial service (e.g., Mobile Money, bank API).
- **Responsibilities:**
 - ✓ Process payments for expert consultations.
 - ✓ Confirm transactions and notify users.

d) Components of a use case diagram

Table 8: Components of a use case diagram

Element	Notation	Description
Actors		Represents an entity that directly interacts with the system. The actor is what performs the different possible actions of the system
Use case		A use case represents a functionality of the system. It is an action that can be performed by an actor.
Association		it indicates that an actor takes part in a use Case
Include		An inclusion denotes that an included action must be performed before the including action can be performed.
Extend		An extension denotes that an extending action may be performed while an extended action is being performed.
Generalization		This shows that an actor or a use case is a kind of another abstract or concrete actors which can be defined and later specialized using generalization relationship.
System		It is a container of use cases which interact with external actors

e) General use case

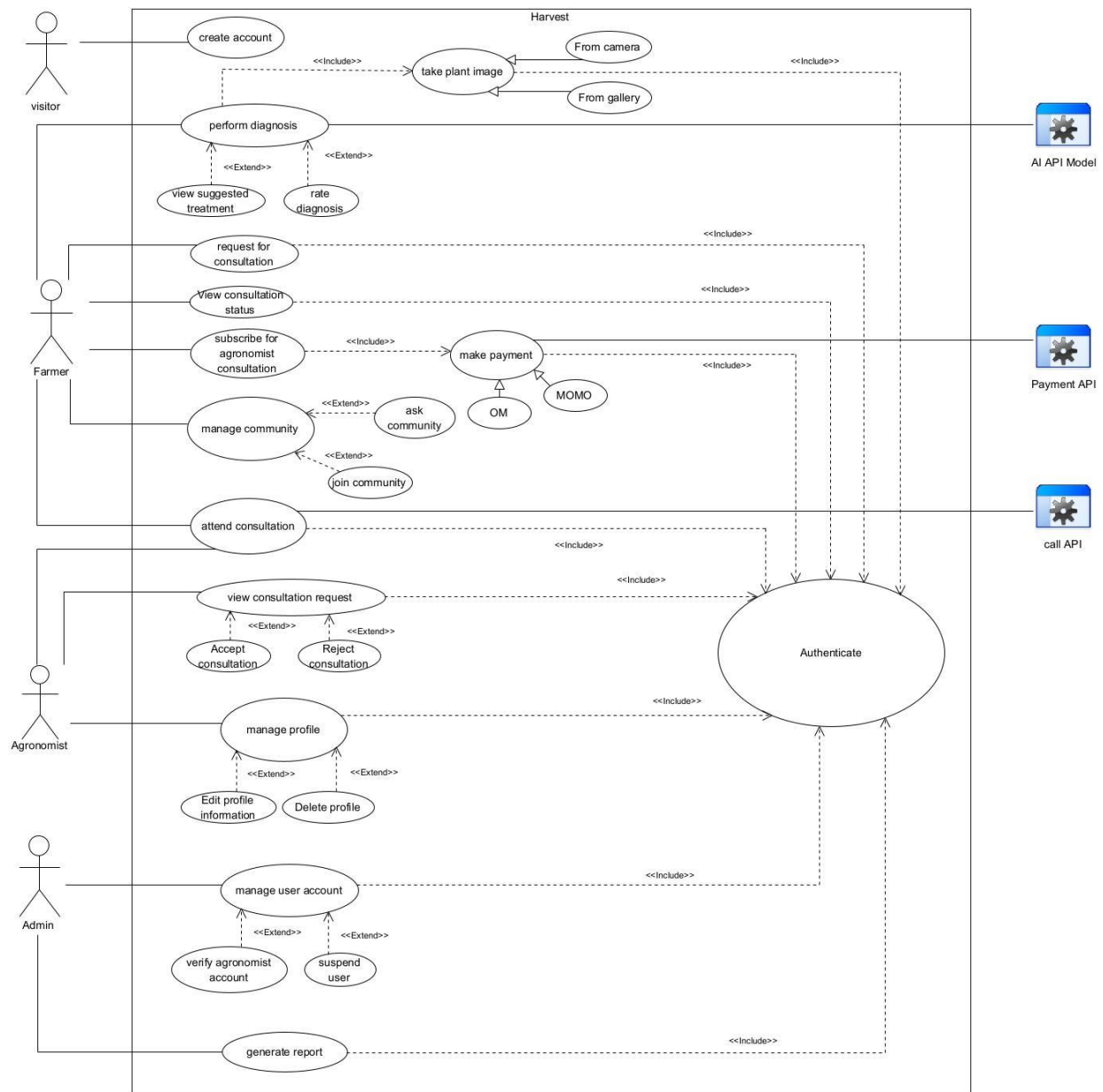


Figure 4: General use case

➤ Specific use case of perform diagnosis

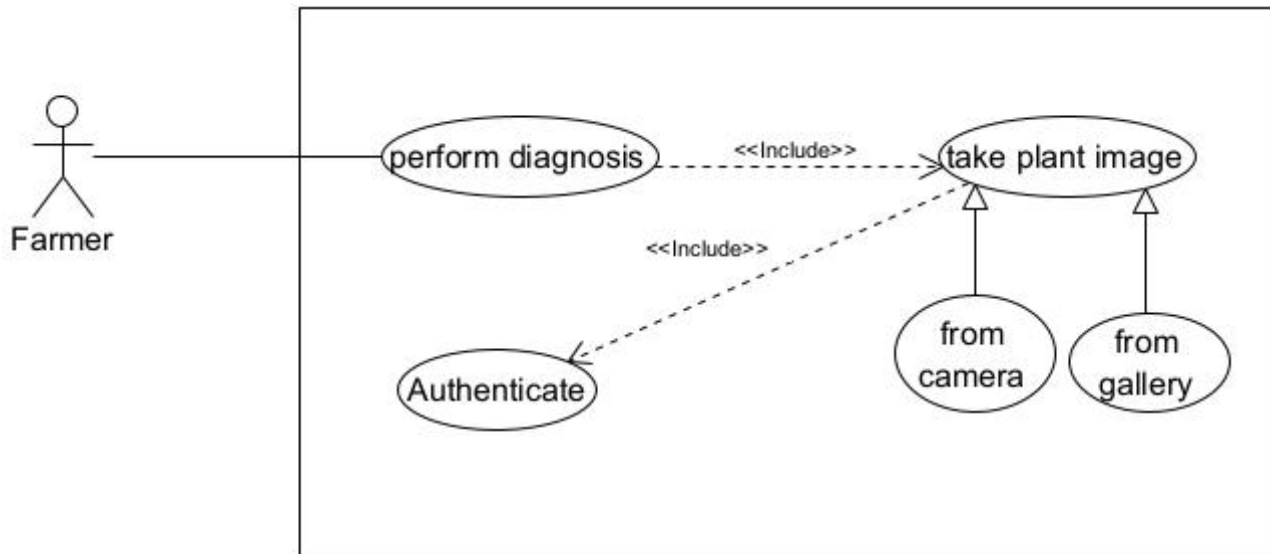


Figure 5: specific use case of perform diagnosis

➤ Specific use case of request for consultation

f) Textual description of use cases

1. Textual description of authenticate use case

Table 9: Textual description of authenticate use case

USE CASE AUTHENTICATE	
Summary	Permit client to access the system
Actors	All user
Pre-condition(s)	- The app is launched and is working perfectly - The user already has an account - The account must be active
Trigger	The user clicks on login button
Nominal scenario	- The user clicks on login button - The system displays the login page - The user fills form and submit - The system checks conformity - The system sends validation query - The system treats the result returned by the database - The user is directed to his page
Alternative scenario	- If step 4 of the nominal scenario the format is not correct, an error message is returned to the user and he returns to step 3 of the nominal scenario. - If step 6 of the nominal scenario no user was found, the system returns an error message and returns to step 3 of the nominal scenario
Post-condition of success	The user logs in successfully and has access to his dashboard
Post condition of failure	An error message is display and user does not have access his dashboard

2. Textual description of request for consultation use case

Table 10: Textual description of request for consultation use case

Title	Textual description of the request for consultation use case
Summary	We request for consultation to an expert to get diagnosis and treatment of our plants
Actor	Farmer
Date	
Stakeholder	EBE OBIANG Orlane
Precondition	- Farmer must authenticate - Farmer must have an active subscription status
Trigger	Navigates to the consult expert page
Nominal Scenario	- Farmer navigates to the consult expert page - System displays a list of experts - Farmer search for an expert, chooses one and clicks on book consultation button - System displays a booking form - Farmer fills and submits the form - System checks for conformity - System sends saving queries to the DBMS - The DBMS executes query and sends a response - System treats the response - System displays a successful message and sends the form to the expert



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Alternative Scenario	11. At step 5, the farmer enters wrong credentials. The system will display an error message 12. At step 8, the DBMS sends a bad response and couldn't save. The system displays an error message
Post-condition of success	The farmer successfully requested for a consultation
Post-condition of failure	The request for consultation failed

3. Textual description of perform diagnosis use case

Table 11: Textual description of perform diagnosis use case

Title	Textual description of the perform diagnosis use case
Summary	We perform diagnosis to know what our plant is suffering of
Actor	Farmer
Date	
Stakeholder	EBE OBIANG Orlane
Precondition	- Farmer must be authenticated
Trigger	Navigate to the diagnose page
Nominal Scenario	1. Farmer navigates to the diagnose page 2. System displays the diagnosis screen 3. Farmer takes or uploads a plant image

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		- Farmer submits the image for diagnosis - System analyses the image format - System sends create diagnosis request to AI model - AI model executes and sends response - System analyses the response - System sends saving query to DBMS - DBMS executes and send response - System displays diagnosis
Alternative Scenario		- At step 3, farmer sends a wrong image format. System displays an error message and displays back the diagnosis screen - At step 7, AI model sends a bad response. System displays an error message - At step 10, DBMS sends a bad response. System displays an error message
Post-condition success	of	The farmer receives the diagnosis
Post-condition failure	of	The farmer doesn't get a diagnosis

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2. COMMUNICATION DIAGRAM

a. Definition

In UML, a communication diagram shows the interactions between the objects or roles associated with lifelines and the messages that pass between lifelines. They describe both the static structure and the dynamic behavior of a system. It is a simplified version of a Collaboration Diagram introduced in UML 2.0.

b. Formalism

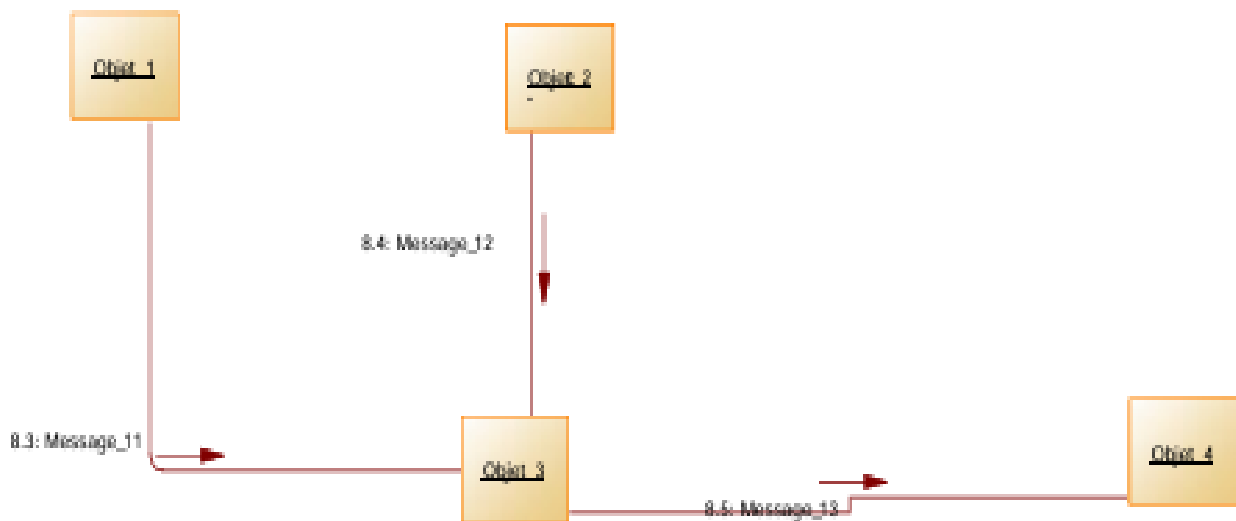


Figure 6: Formalism of communication diagram

c. Components of a communication diagram

Table 12: Component of a communication diagram

Element	Description and Main properties	Notation
Call message	A call message defines a particular communication between lifelines of interaction that represents an innovation of target lifeline.	
Dependency		
	A dependency is a relationship that signifies that a single or set of model elements require other model elements for their specification.	2: dependency 
Object		
	An object represents an individual participant in the interaction conversation	
Generalization	A generalization is a taxonomic relationship between a more general classifier and a more specific classifier	

3. SEQUENCE DIAGRAM

a. Definition

A sequence diagram in UML (Unified Modeling Language) is a type of interaction diagram that shows the sequence of events or interactions between objects or components in a system. It illustrates how objects communicate with each other,. These can help to predict how a system will behave and to discover responsibilities a class may need to have in the process of modelling a new system.

b. Formalism

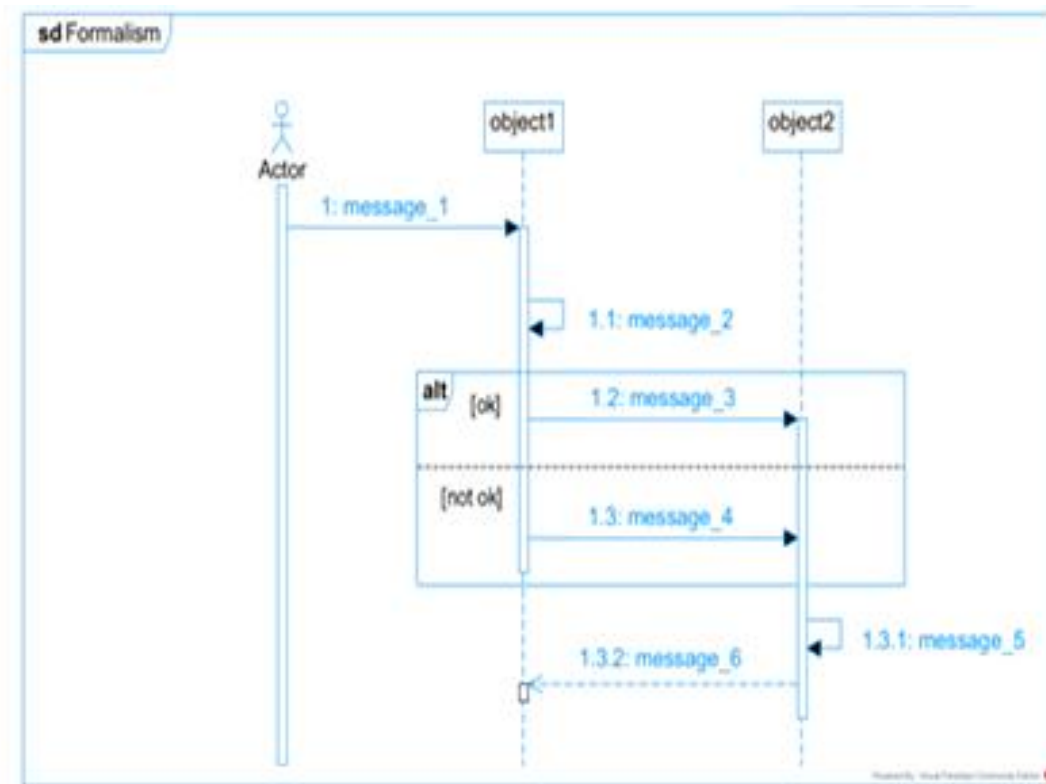
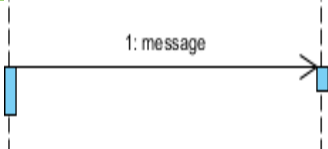
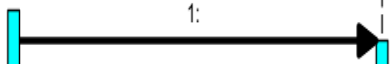
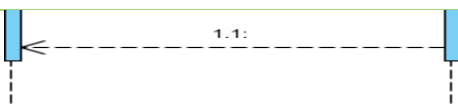


Figure 7: Formalism of a sequence diagram

C) Components of a sequence diagram

Table 13: Components of a sequence diagram

Element	Notation	Description
Lifelines		They represent rows or objects instances that participate in the sequence being modelled.
Asynchronous Message		It is a message that receives an indirect response.
Synchronous Message		It is a message that sends and want response before it continues a process
Self-Message		Self-message is a kind of message that represents the invocation of message of the same lifeline.
Return Message		It represents the response of a message.
Actor		They send and receive message.
Combined Interaction Fragment		An articulation of interaction diagram, defined by an operator and operands.
Object		They send and receive messages

d) Sequence diagram of perform diagnosis

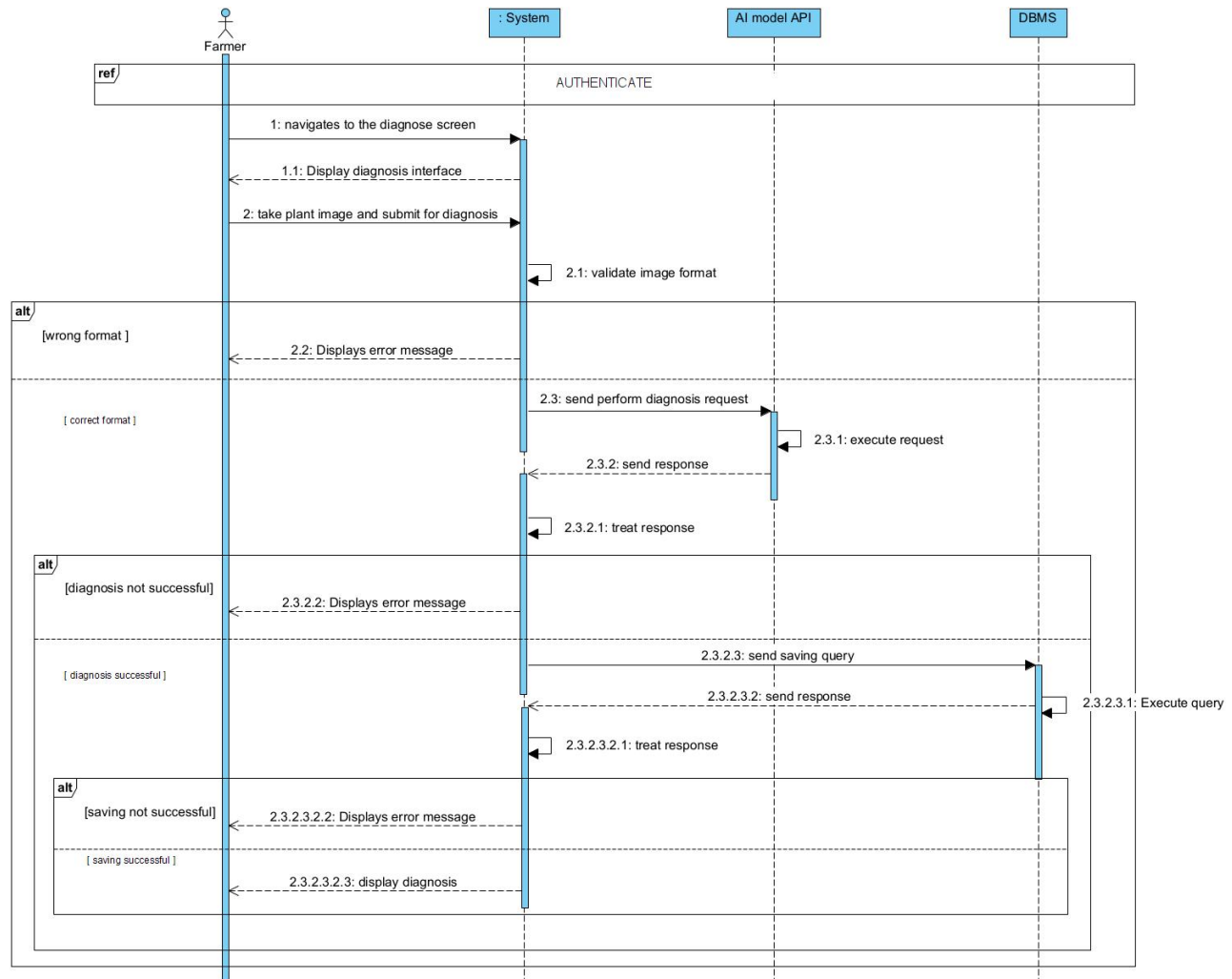


Figure 8: sequence diagram of perform diagnosis

e) Sequence diagram of request for consultation

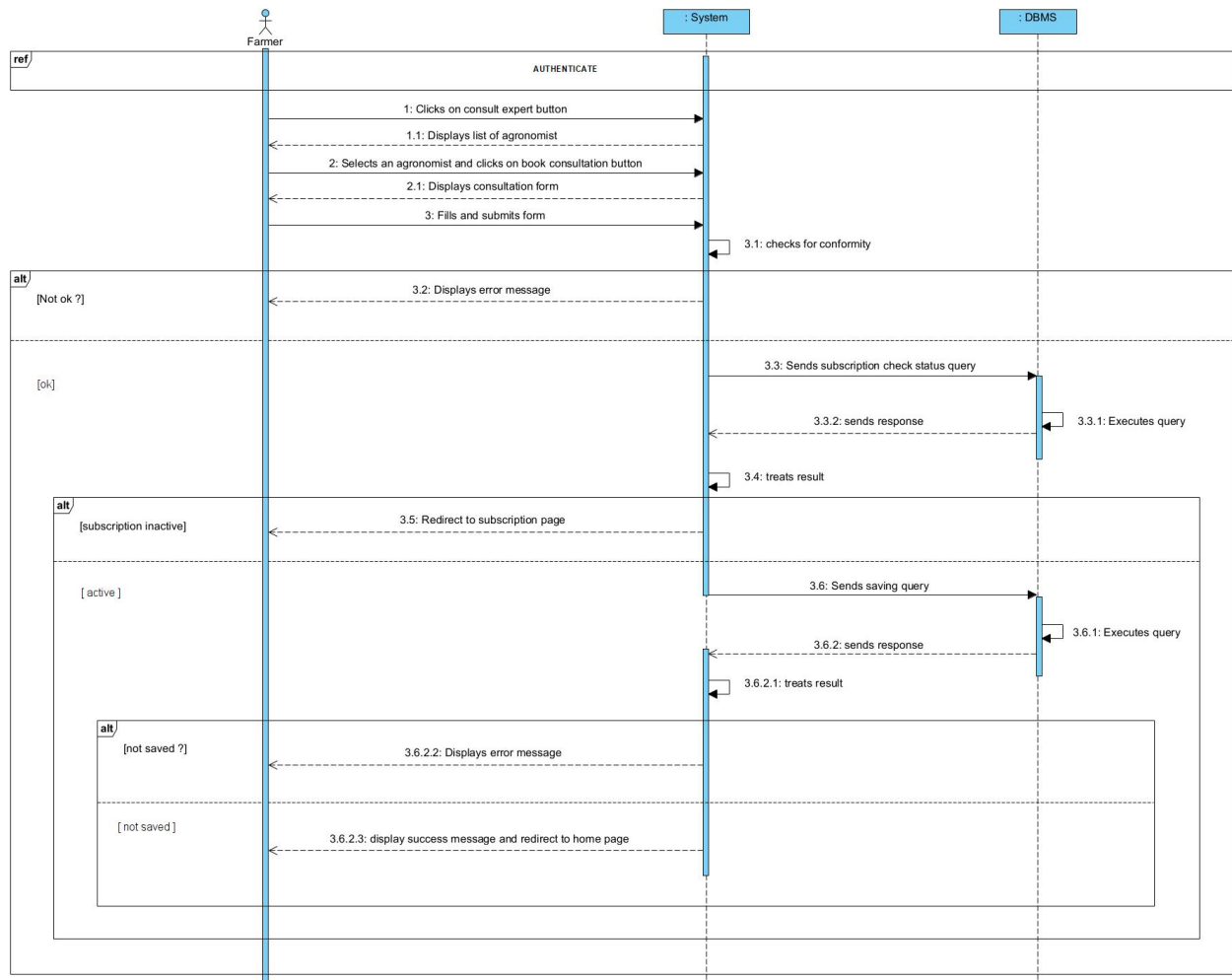


Figure 9: Sequence diagram of consult expert

4. ACTIVITY DIAGRAM

a. Definition

An activity diagram is a graphical representation of workflows that show the steps needed in the realization of a process; showing the details from a start point to an end point through all decisions and actions that can possibly be performed. Activity diagrams are intended to model both the computational and organizational process. They flow can be sequential, branched or concurrent. Below is an activity diagram formalism.

b. Formalism

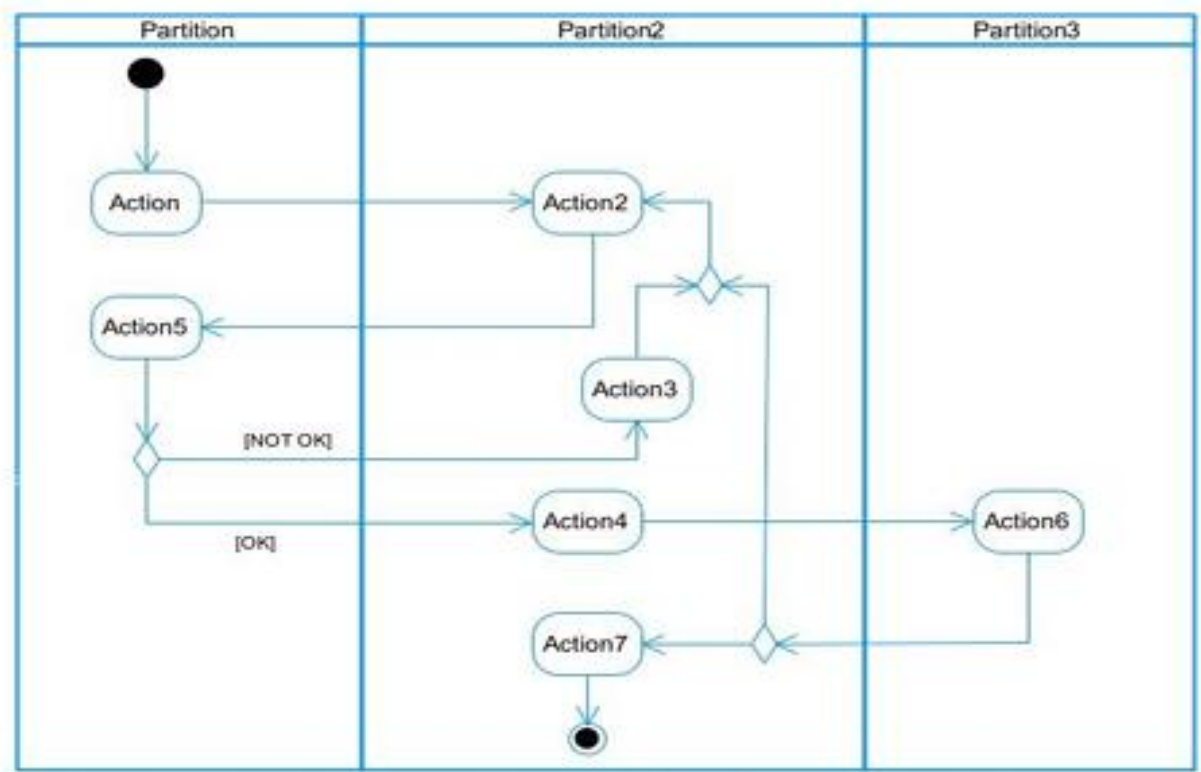
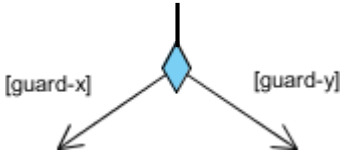
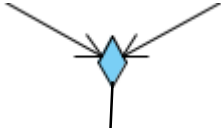


Figure 10: Formalism of an Activity diagram

c) Components of an activity diagram

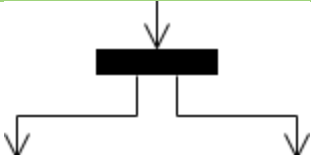
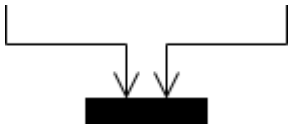
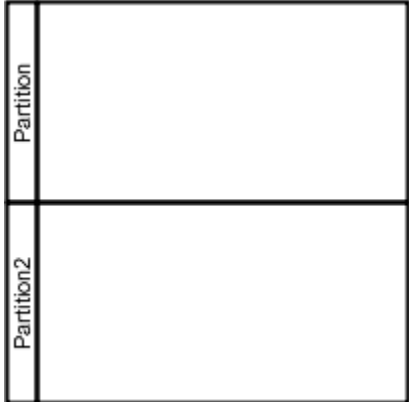
Table 14: Components of an activity diagram

Element	Notation	Description
Activity		Is used to represent a set of actions
Action		A task to be performed
Activity Edge		An activity edge in a UML activity diagram is a directed connection between two activity nodes. It represents the flow of control or data between these nodes.
Initial node		Portrays the beginning of a set of actions or activities
Activity Final Node		Stop all control flows and object flows in an activity (or action)
Object node		Represent an object that is connected to a set of Object Flows
Decision node		Represent a test condition to ensure that the control flow or object flow only goes down one path
Merge node		Bring back together different decision paths that were created using a decision-node.



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Fork node		Split behavior into a set of parallel or concurrent flows of activities (or actions)
Join node		Bring back together a set of parallel or concurrent flows of activities (or actions).
Swimlane and Partition		A way to group activities performed by the same actor on an activity diagram or to group activities in a single thread

d) Activity diagram of perform diagnosis

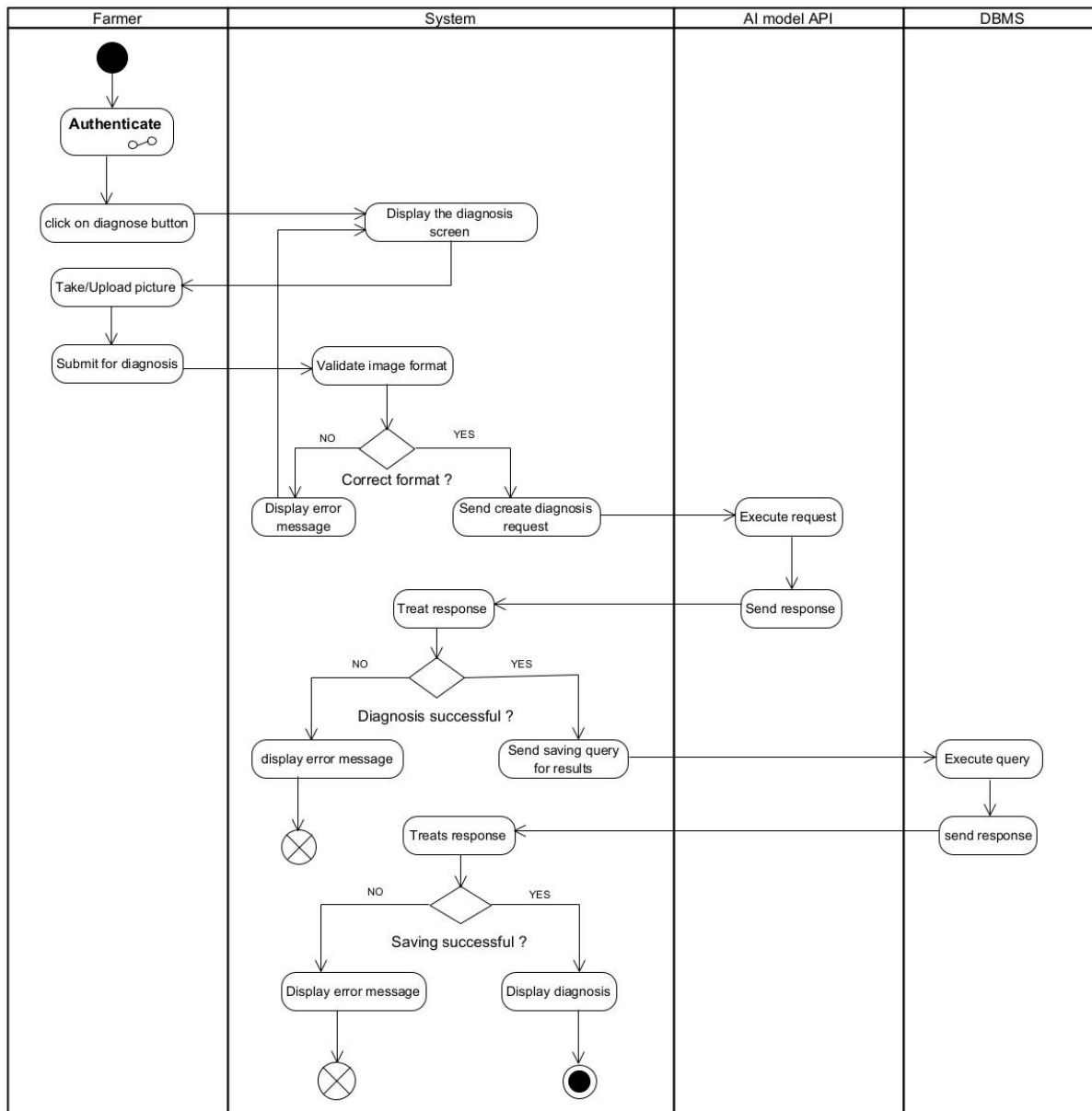


Figure 11: Activity diagram of perform diagnosis

e) Activity diagram of request for consultation

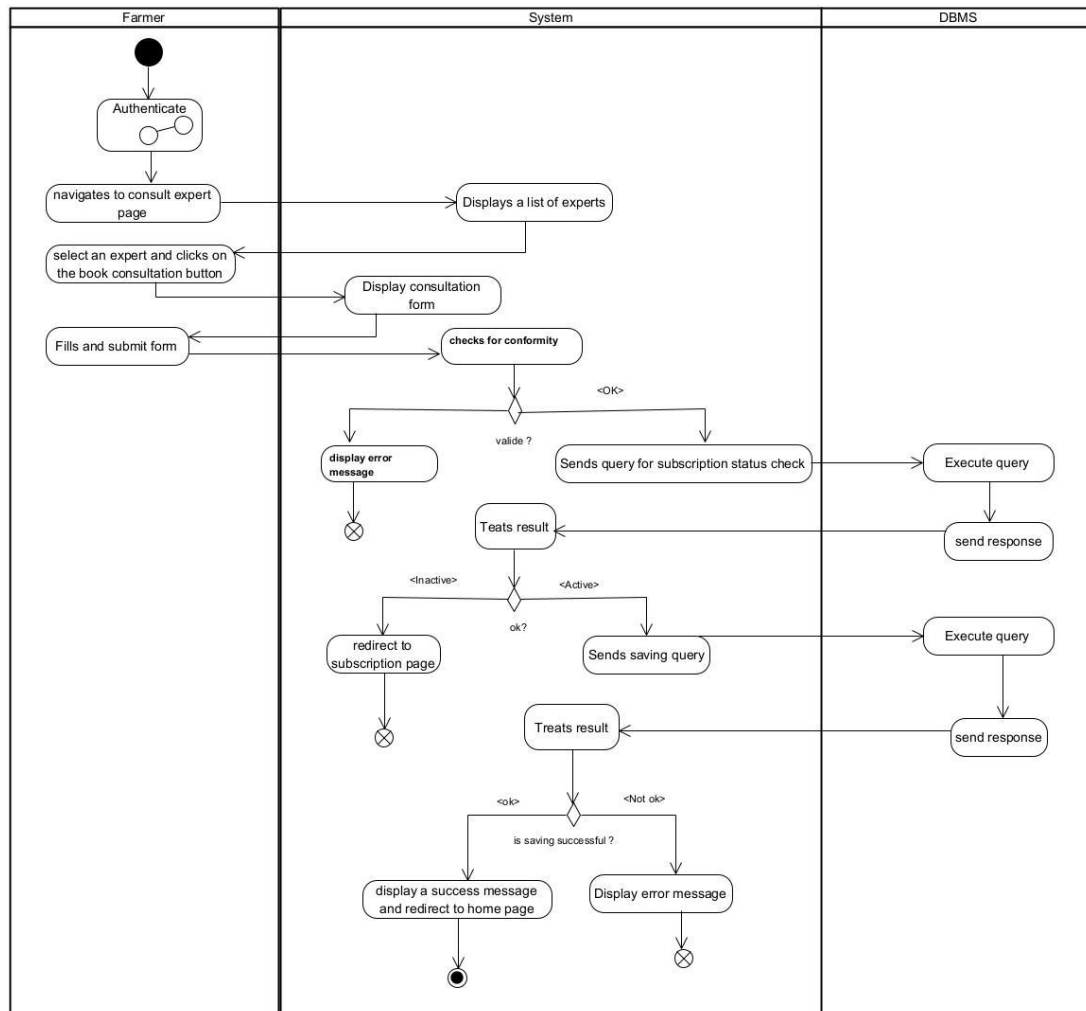


Figure 12: Activity diagram of request for consultation



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CONCLUSION

In the analysis phase, we chose a software development process and modelling language, after which we explained the functional need of our system, we saw the use case diagram which shows the relationship between the actors and use case (the action the actor can perform on the system), we saw the communication diagram which represents the architecture of the system based on object oriented programming, we saw the sequence diagram which represents the flow of messages between elements in the system, and lastly the activity diagram which shows the workflow of our system. We will now move to the conception phase in which we will present the Technical branch of our system together with related diagrams.



PART IV: CONCEPTION PHASE



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Preamble

The conception phase will permit us to present in an orderly manner the components necessary for the good functioning of our software and also the architecture used for the proposed solution. It bridges the gap between the analysis phase and the realization phase.

Content

INTRODUCTION

- I. CLASS DIAGRAM
- II. STATE MACHINE DIAGRAM
- III. PACKAGE DIAGRAM

CONCLUSION



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INTRODUCTION

The conceptual phase will describe in details the necessary specifications, features and operations that will satisfy the functioning requirements of the proposed system as modelled in the analysis phase. This phase is meant to identify and consider essential components (hardware /or software), structure (network capabilities), processes and procedures for the system to accomplish its objectives. We will look at some diagrams such as the component diagram, package diagram, communication diagram, class diagram and the object diagram.

I. CLASS DIAGRAM

a. Definition

A class diagram is a static diagram. It represents the static view of an application. class diagram is not only used for visualizing, describing and documenting different aspect of the system but also for constructing executable code of the software application. Class diagram describes the attribute and operation of a class and constraints imposed on the system. Its purpose is to model the static view of an application.

b. Formalism

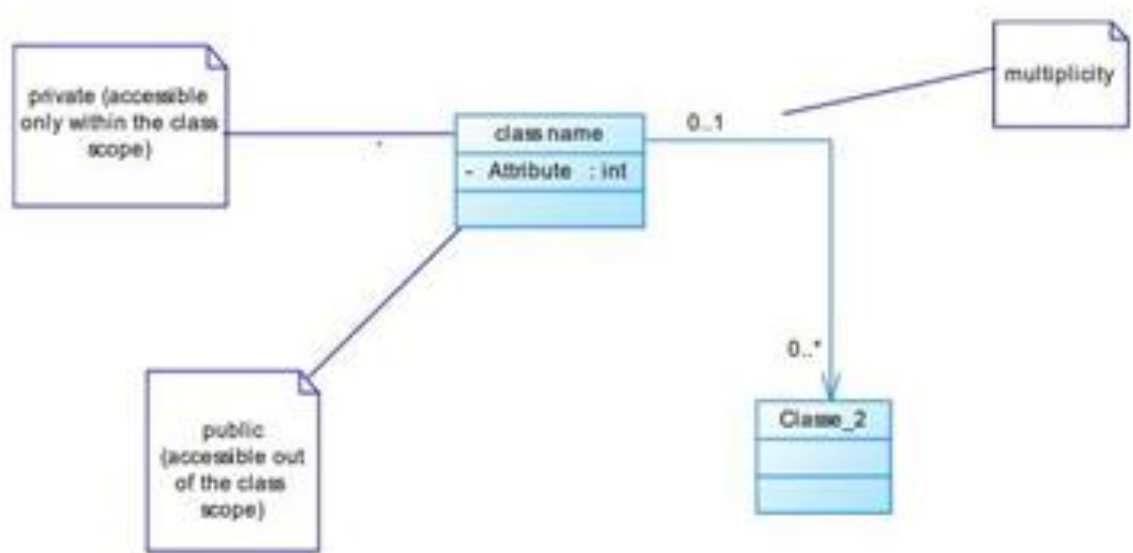
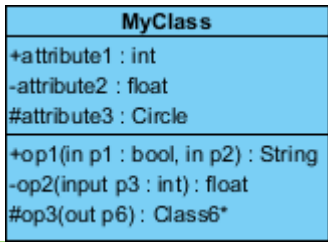
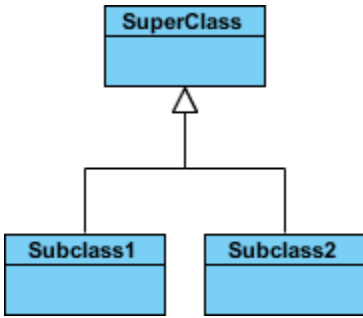
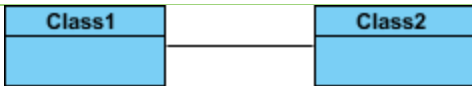
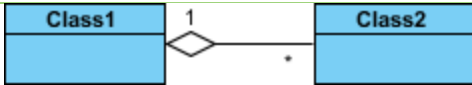
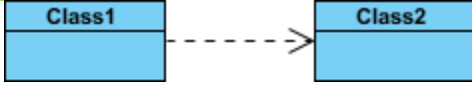


Figure 13: Formalism of a class diagram

c) Components of a class diagram

Table 15: Components of a class diagram

Element	Notation	Description
Class	 <pre> classDiagram class MyClass { +attribute1 : int -attribute2 : float #attribute3 : Circle +op1(in p1 : bool, in p2) : String -op2(input p3 : int) : float #op3(out p6) : Class6* } </pre>	A class is an element that defines the attributes and behaviors that an object can generate
Generalization	 <pre> classDiagram SuperClass < -- Subclass1 SuperClass < -- Subclass2 </pre>	it a relationship between a whole thing (called superclass) and a more specific thing (called subclass)
Association	 <pre> classDiagram Class1 -- Class2 </pre>	It is a general type of relationship between elements, it may include cardinality, roles etc.
Aggregation	 <pre> classDiagram Class1 o-- "*" Class2 </pre>	If the parent of the aggregate is deleted, the children are not deleted.
Composition	 <pre> classDiagram Class1 *-- "*" Class2 </pre>	If a parent of a composite is deleted all its parts are deleted with it.
Dependency	 <pre> classDiagram Class1 ..> Class2 </pre>	It existed between two classes, if one changes it may cause the change in the order, but the other way around

d) Class diagram

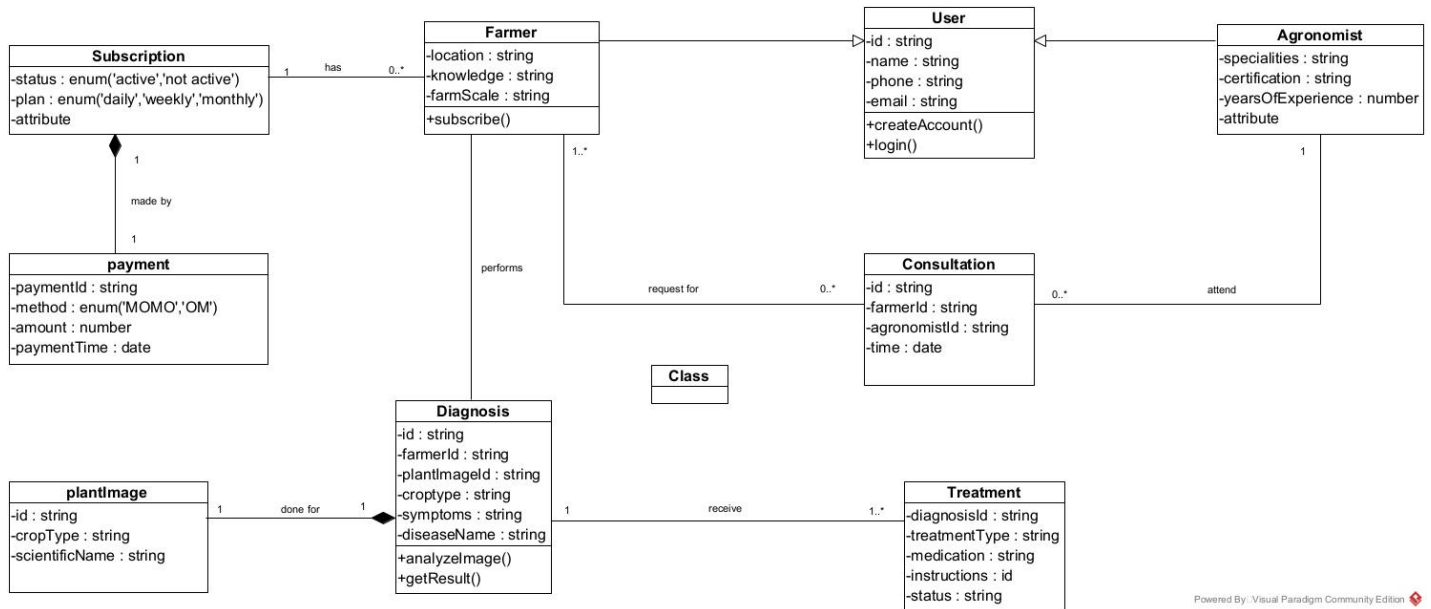


Figure 14: Class diagram

II. STATE MACHINE DIAGRAM

a. Definition

A state machine diagram describes the behavior of a single object in response to a series of events in a system. Also known as the state machine diagram, it models the dynamic flow of control from the state of a particular object within a system.

b. Formalism

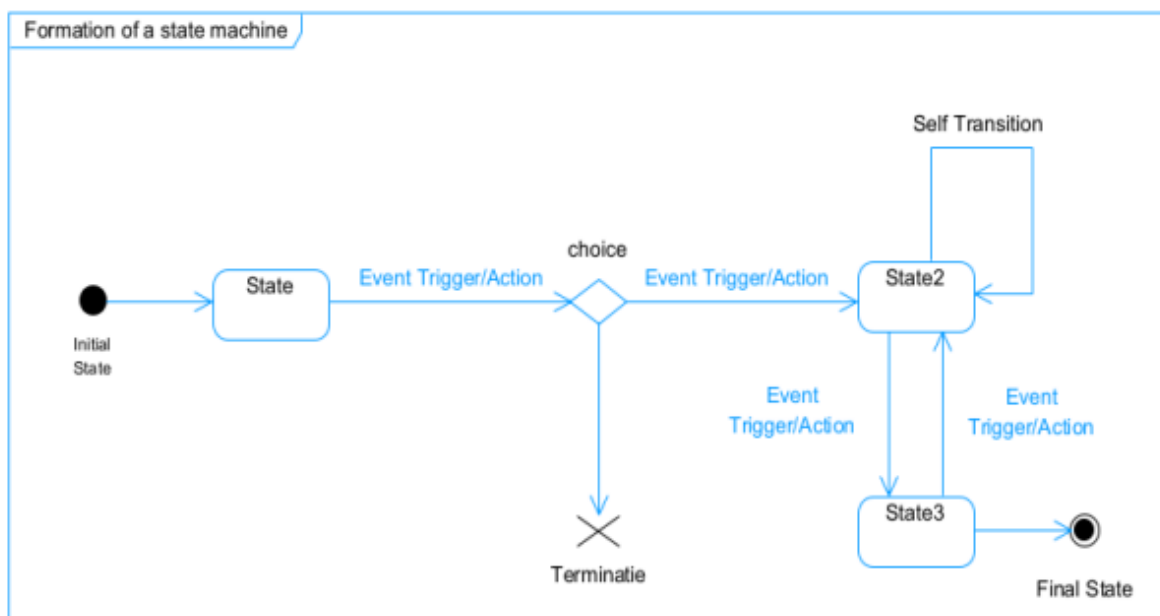


Figure 15: Formalism of a state machine diagram

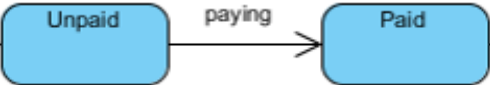
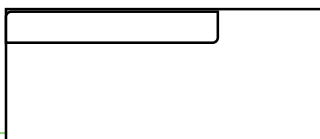


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c) Components of a state machine diagram

Table 16: Components of a state machine diagram

Element	Notation	Description
State		Models a situation during which a certain invariant condition holds.
First (Initial State)		It represents a default vertex, that is, a source for a single transaction to the default or composite state.
Final state		A state specifying that the enclosing region is complete
Transition		A direction relation between a source and a target vertex.
Choice pseudo state		A diamond symbol that indicates a dynamic condition with branched potential results
Terminate		Implies that the execution of a state by means of its context is terminated.
Diagram overview		A placeholder for the linked states in a state machine diagram.



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d) State machine diagram of expert's account

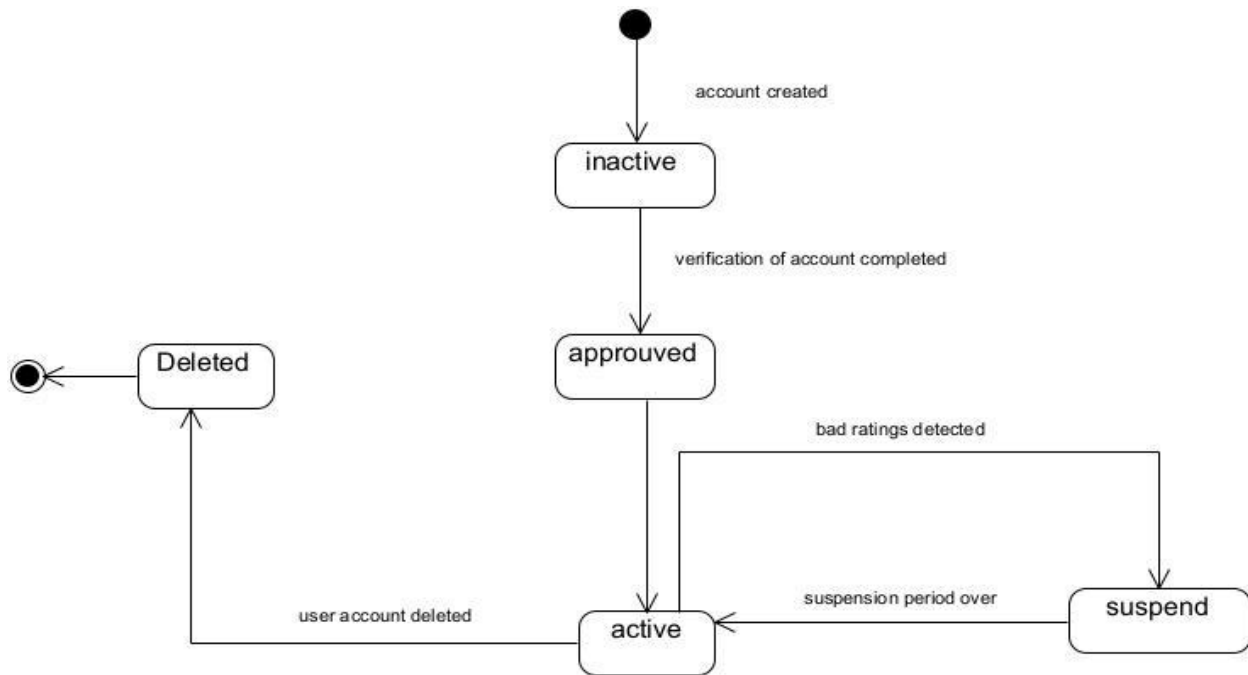


Figure 16: State machine of expert's account

e) State machine of subscription status of farmer

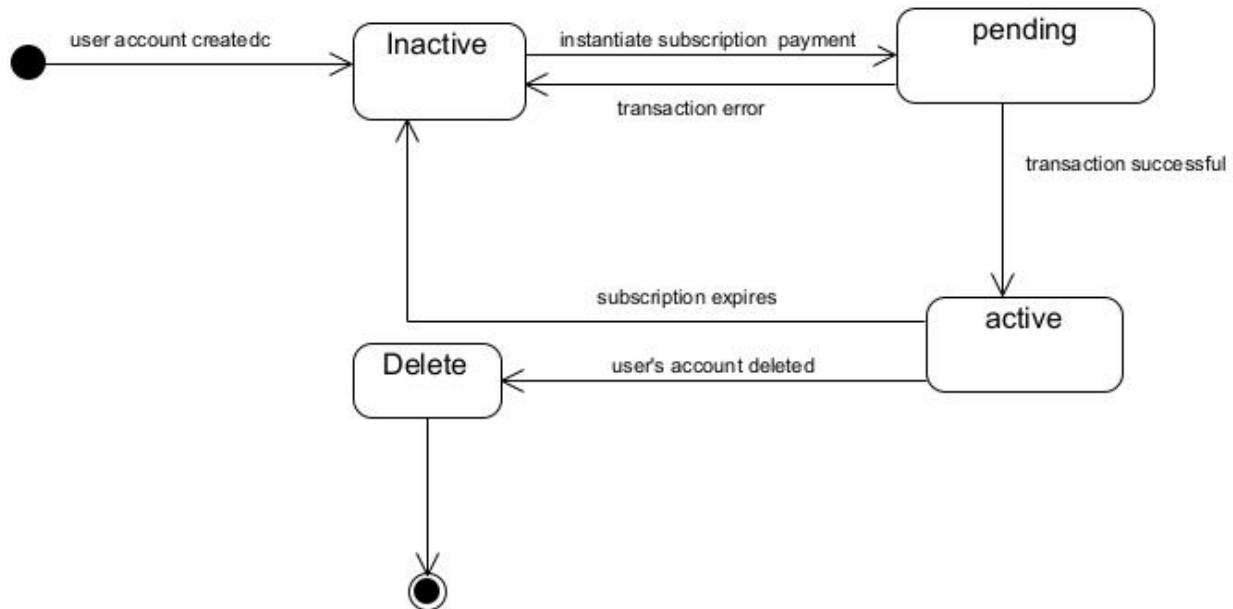


Figure 17: State machine of subscription status of the expert

III. PACKAGE DIAGRAM

a. Definition

This is a structural diagram used to show the organization and arrangement of various model elements in the form of packages. A package diagram is the grouping of related uml elements such as classes, diagrams or even other packages.

b. Formalism

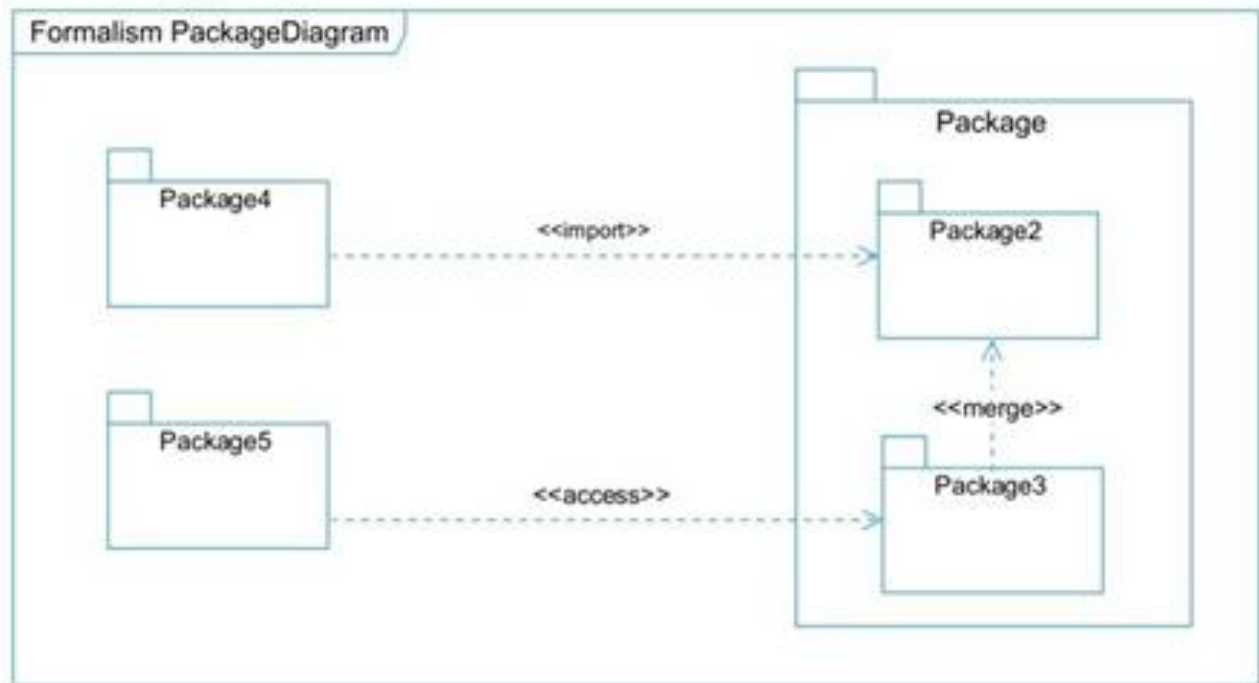
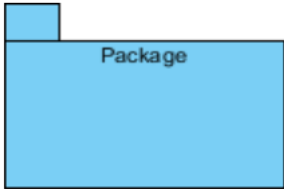
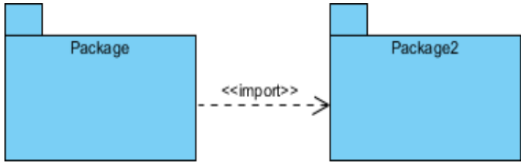
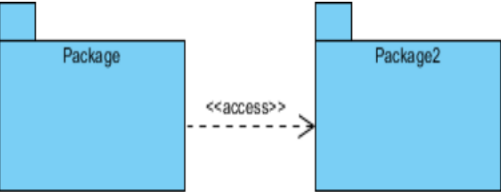


Figure 18: Formalism of a package diagram

c) Components of a package diagram

Table 17: Components of a package diagram

Element	Notation	Description
Package		A package is a namespace use to group related elements; it is a mechanism used to group elements into a better structure in a system.
Package import		A relationship Indicate that, functionality has been imported from one package to another.
Package access		A relationship Indicates that one package requires assistance from the function of another package.
Package merge		It is a relationship which shows that, the functionality of two packages are combines to a single function.

d) Package diagram

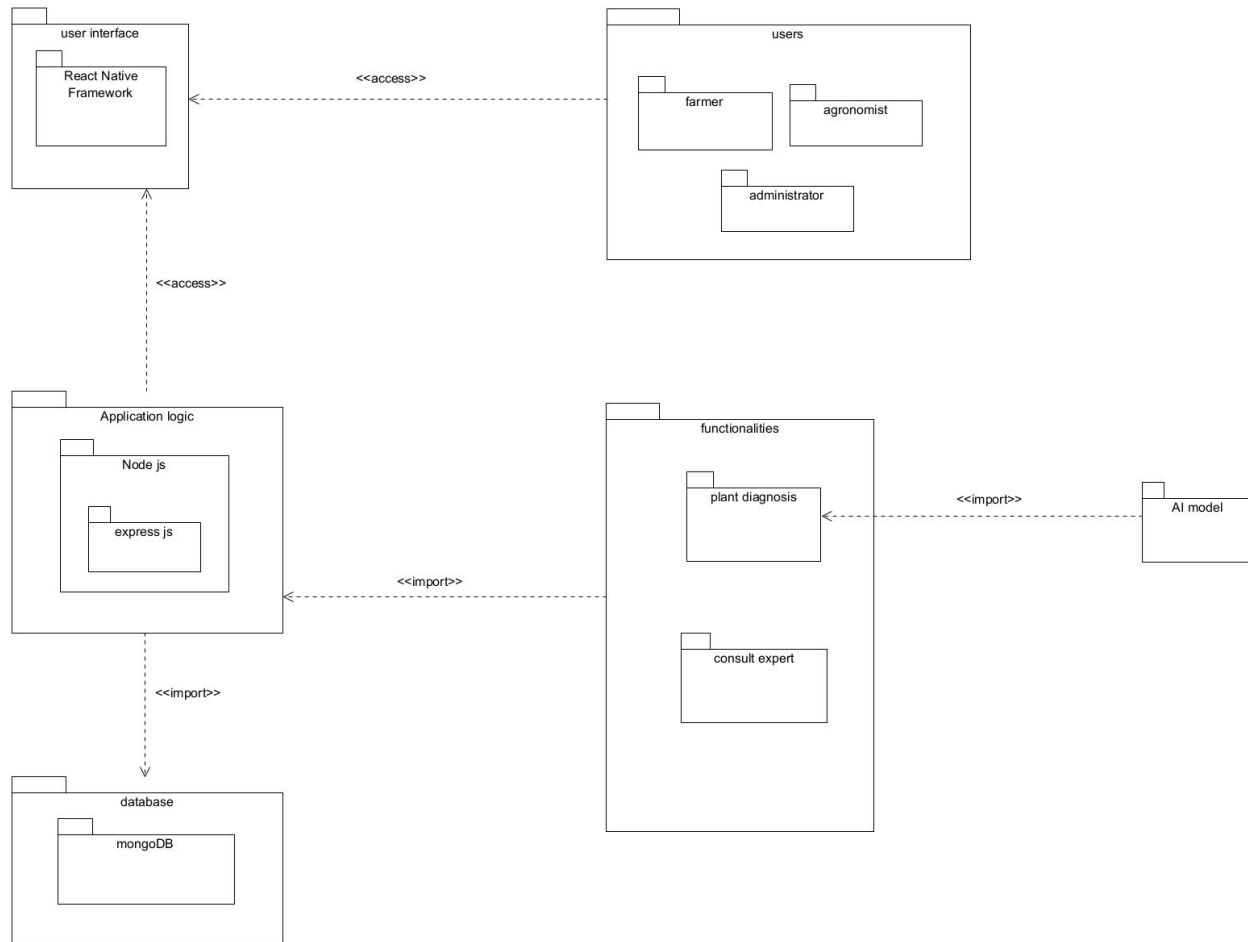


Figure 19: Package diagram



CONCLUSION

In the conception phase we set the stage for a successful project through the creation of key diagrams such as class diagrams, state machine diagrams, and package diagrams. This phase facilitates a shared understanding among stakeholders, identifies and mitigates potential risks and issues, and establishes a robust architectural foundation for implementation. We finished this phase by looking at interaction between the various aspects and actors of our system. The next phase of our report is the realization phase where we will look at aspects concerning the implementation of our system.



PART V: THE REALIZATION PHASE

Preamble

In this phase we will to straight forward in the implementation of our solution, we will base ourselves on the analysis and conception phases and also present the component and deployment diagrams.

Content

INTRODUCTION

I. ARCHITECTURE USED

1. PHYSICAL ARCHITECTURE

2. LOGICAL ARCHITECTURE

II. DEPLOYMENT DIAGRAM

III. COMPONENT DIAGRAM

CONCLUSION



IMPLEMENTATION OF A MOBILE APPLICATION FOR PLANT DIAGNOSIS



INTRODUCTION

In the realization phase, we will talk about the architecture we use and some diagrams related to the physical aspect of the system like libraries, documents, as well as the physical topology of the components of the system when the software is been deployed.

I. ARCHITECTURE USED

1. PHYSICAL ARCHITECTURE

The design of the DBMS depends on its architecture. An n-tier architecture partitions on the whole system into related but separated n modules, which can be independently modified, altered, changed or replace. A large amount of data on web servers, personal computers (pc) and others are link with networks with the help of basic client or server architecture. Within the scope of our project, we made use of the n-tier architecture this architecture separates it tiers from each other based upon the user and the manipulated data in the database. Each layer has a well-defined communication interface, and the evolution of the layer is independent of the other. The n-tier of our system is made up of:

- ❖ **The hardware tier**, which represents our IOT device to collect vitals and it is connected directly to our mobile phone using a wired connection.
- ❖ **The client tier**, which is also known as our presentation interface.
- ❖ **Application Tier**, which represents our webserver. The data tier, which represents our DBMS server.

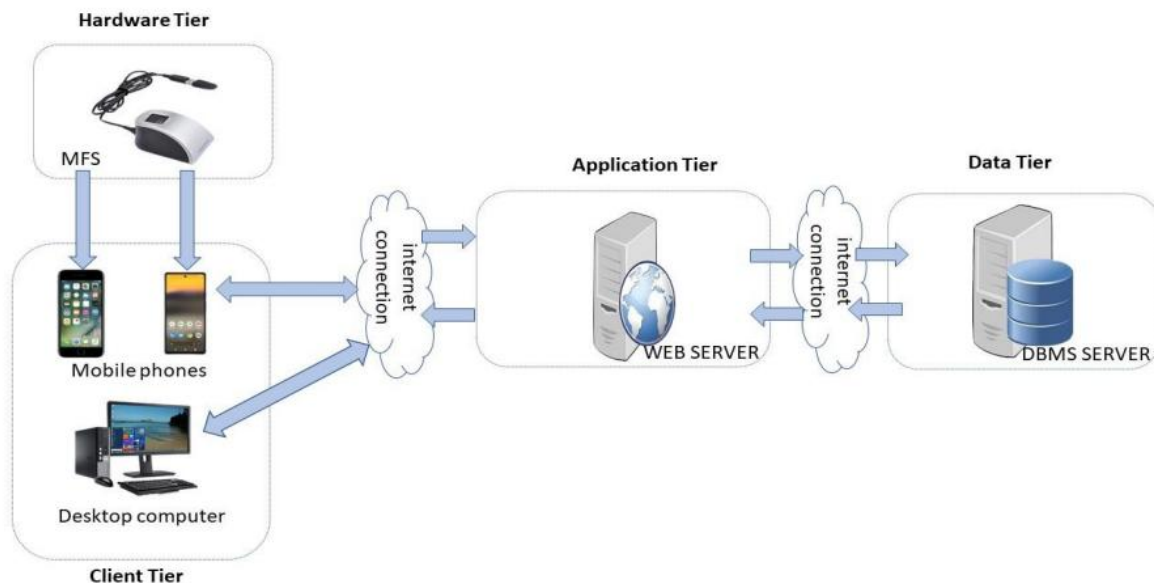


Figure 20: n-tier Architecture (Source: <https://www.pinterest.es/pin/752241943987565648/>)

2. LOGICAL ARCHITECTURE

The Model-View-Controller (MVC) framework is an architectural/design pattern that separates an application into three main logical components Model, View, and Controller. Each architectural component is built to handle specific development aspects of an application: Here's what each of those components mean:

- **Model:** The backend that contains all the data logic
- **View:** The frontend or graphical user interface (GUI)
- **Controller:** The brains of the application that controls how data is displayed

MVC Architecture Pattern

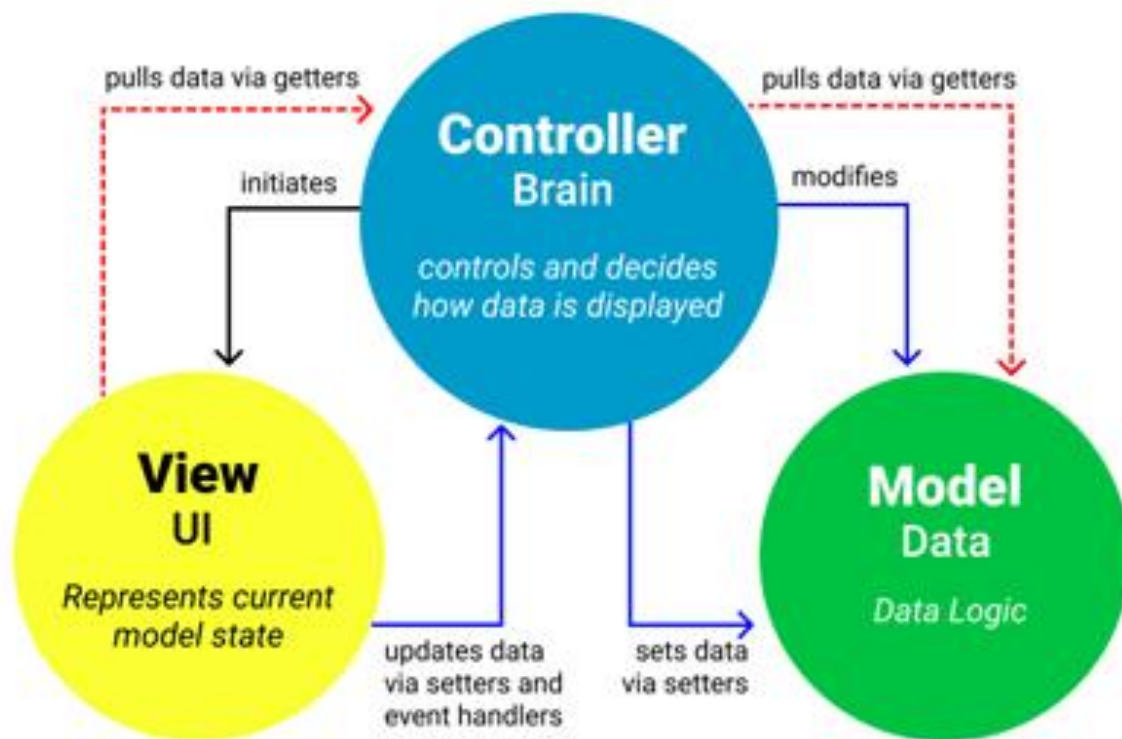


Figure 21: The MVC architecture (source: <https://www.freecodecamp.org/news/the-model-view-controller-pattern-mvc-architecture-and-frameworks-explained/>)

II. DEPLOYMENT DIAGRAM

a. Definition

Deployment diagram is a structural diagram used to visualize the topology of the physical components of a system, where the software is deployed. They consist of nodes and their relationship. It is related to the component diagram because the components are deployed using the deployment diagram. A deployment diagram consists of nodes (physical hardware used to deploy the application.).

b. Formalism

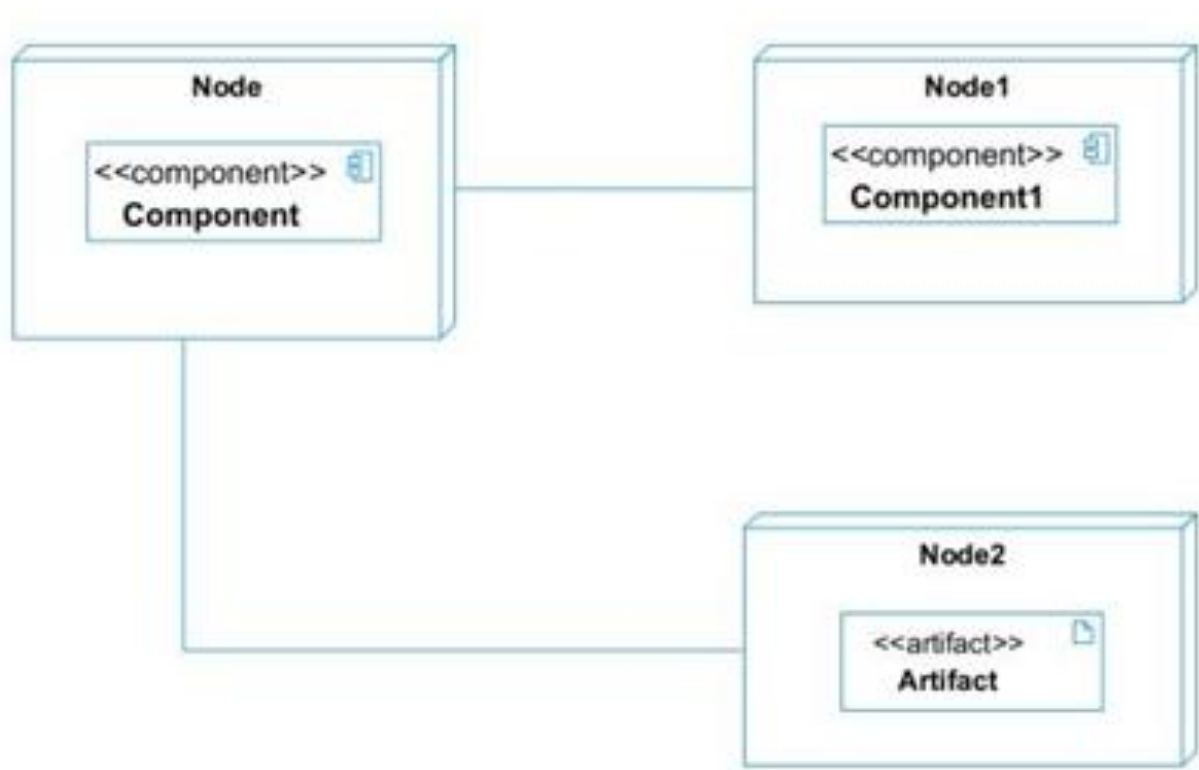
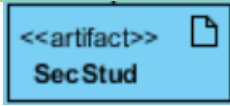
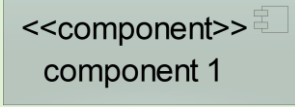


Figure 22: Formalism of deployment diagram

c) Deployment diagram

Table 18: Components of a deployment diagram

Element	Notation	Description
Node		A hardware or software object, shown by a three-dimensional box
Artifact		An artifact is a major product, which is produced or used during the development of a software. E.g diagrams, data models, setup scripts
Component		It represents a modular part of a system that encapsulates its content and whose manifestation is replaceable within its environment.
Association		An association helps to connect two nodes together which permits them to communicate together



IMPLEMENTATION OF A MOBILE APPLICATION FOR PLANT DIAGNOSIS



d) Deployment diagram

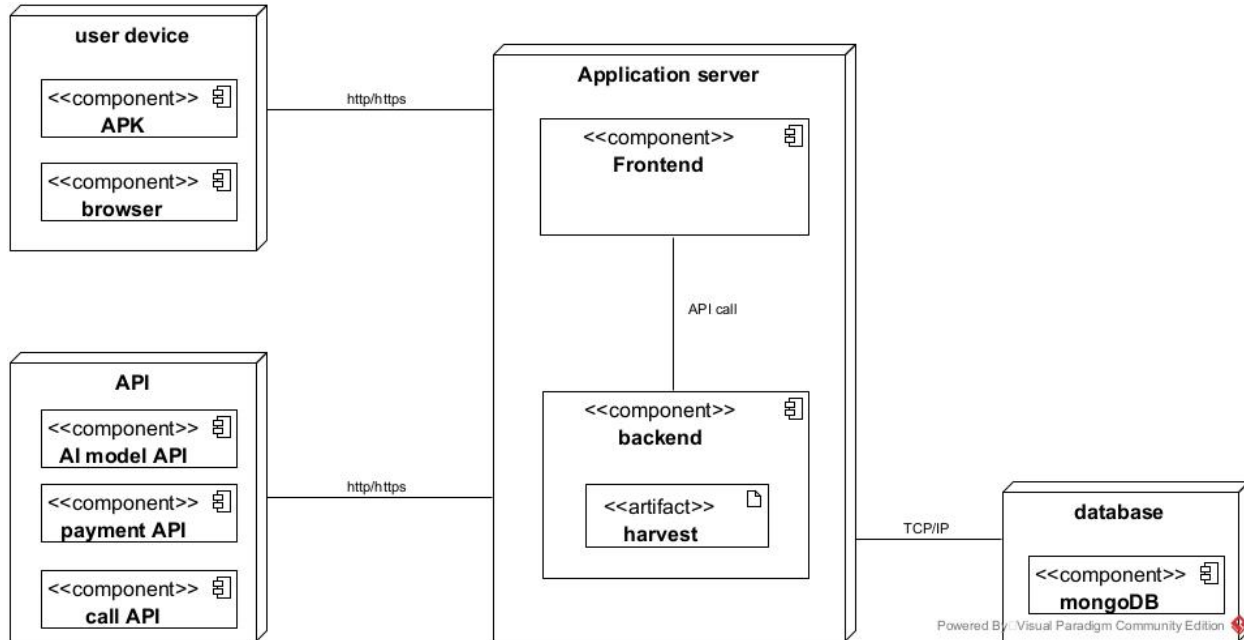


Figure 23: Deployment diagram

III. COMPONENT DIAGRAM

a. Definition

Component diagrams are used to model the physical aspect of a system. Now the question is what are this physical aspect? They are elements such as Executables, libraries, files, document etc. which resides in a node. The component diagram does not describe the functionality of the system, but it describes the components used to make those functionalities.

b. Formalism

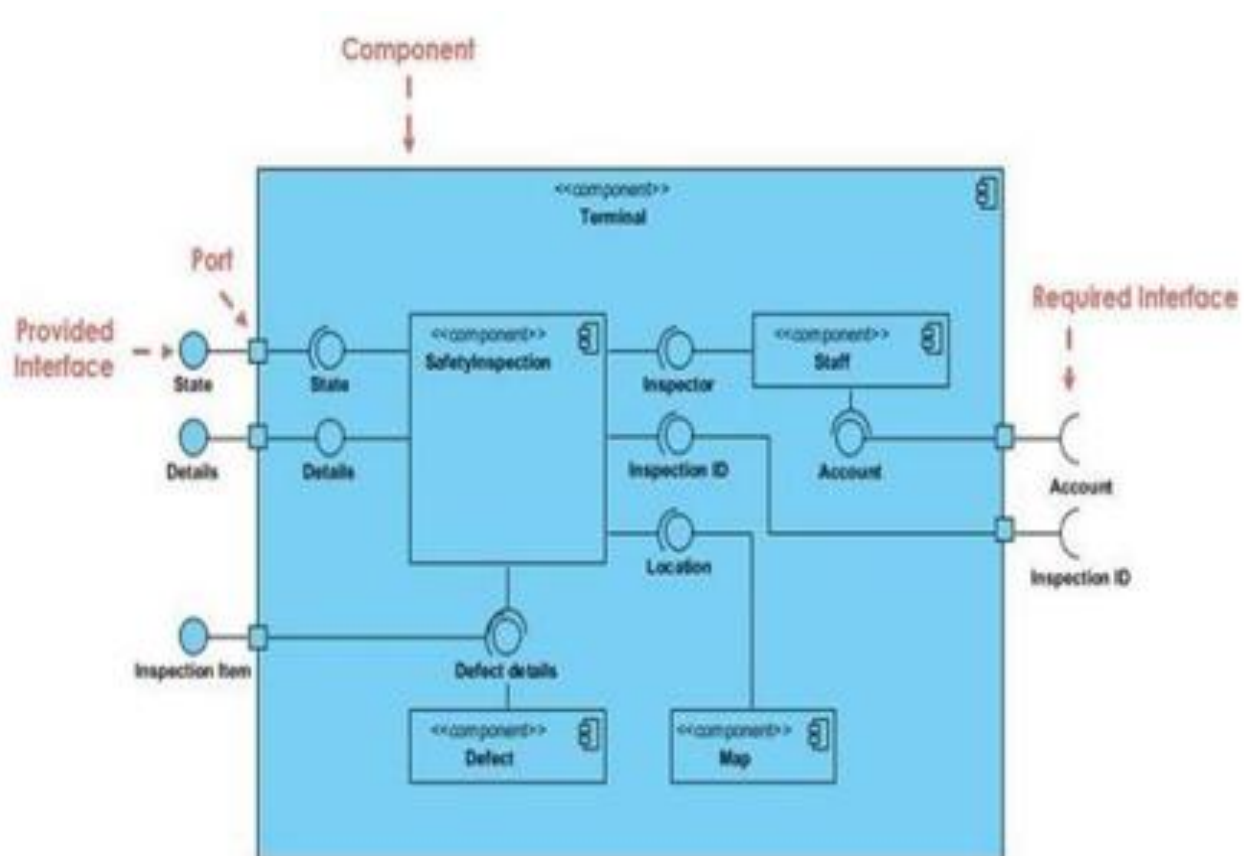


Figure 24: Formalism of component diagram

c) Components of a component diagram

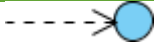
Element	Notation	Description
Dependency		A dependency is a relationship that signifies that a single or a set of model elements requires other model elements for their specification or implementation.
Port		A port is often used to help expose required and provided interfaces of a component.
Interface		It is a point of interaction between a component and its environment or other components. It specifies the services that a component provides or requires from other components
Component		It represents a self-contained and replaceable unit that interacts with other components via provided and required interfaces

Figure 25: Components of a component diagram



IMPLEMENTATION OF A MOBILE APPLICATION FOR PLANT DIAGNOSIS



CONCLUSION

In our realization phase, we implemented our application. In other to accomplish this phase, we made use of our analysis and conception phase. We also drew the deployment and component diagrams which depict the structure of our system in terms of modules, files, assets, how the different elements interact with each other. We will move to the test of functionalities phase, where we will examine the different modules present in our app and how beneficial they are to its different users.



PART V: TEST OF FUNCTIONALITIES



IMPLEMENTATION OF A MOBILE APPLICATION FOR PLANT DIAGNOSIS



INTRODUCTION

The test of functionalities phase helps us to know more about the solution we are building. It provides the different functionalities or modules found in our application and how they are beneficial to the users. Hence, in this phase we are going to explore the different functionalities present in our application.



I- APPLICATION FUNCTIONALITIES

1. Farmer Functionalities

- Secure account creation and login
- Take a picture of plant
- Perform a diagnosis with the AI
- Book consultation from an expert
- Join community of farmers
- Attend consultation with the expert

2. Expert Functionalities

- Register and log in to a professional account
- Create and edit profile
- Manage consultation requests from farmers
- Attend consultation with farmer

3. Administrator Functionalities

- Secure login to the admin dashboard
- Manage users (farmer and expert)
- Suspend or delete fraudulent or bad rated expert accounts

4. Advanced (Optional/Future) Features

- AI-powered epidemic alerts



II. UNIT TESTING

2. What is Unit Testing

Unit testing involves testing the smallest parts of an application, such as functions or methods, independently from the rest of the application. The primary goal is to validate that each unit of the software performs as expected.

2. Unit Testing Importance

Unit testing ensures that each component of the application works independently and correctly. It helps catch bugs early in the development process and guarantees that future changes don't break existing functionality.

3. Testing Strategy

We implemented **unit testing** to isolate and test individual parts of the app, specifically the API endpoints and backend services. Unit testing provides fast feedback on code quality and behavior.

4. Tools Used

- **Jest:** Testing framework used for running unit tests.
- **Supertest:** For testing API requests.
- **Mockingoose:** Used in some instances to mock MongoDB models for tests.

5. Key Functionalities Tested

a. User Registration

- Tests ensure that users can successfully register with valid details.
- Handles test cases where invalid inputs or missing fields cause registration failure.

b. User Authentication

- Tests confirm that users can log in with correct credentials.
- Includes scenarios where login fails due to invalid credentials (e.g., wrong password).



IMPLEMENTATION OF A MOBILE APPLICATION FOR PLANT DIAGNOSIS



CONCLUSION



PART VI: USER GUIDE



IMPLEMENTATION OF A MOBILE APPLICATION FOR PLANT DIAGNOSIS



I. INSTALLATION OF THE APPLICATION



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II. screenshots of the web application



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IMPLEMENTATION OF A MOBILE APPLICATION FOR PLANT DIAGNOSIS



CONCLUSION



IMPLEMENTATION OF A MOBILE APPLICATION FOR PLANT DIAGNOSIS



PERSPECTIVES

This prototype is just the beginning and honestly, that's the most exciting part. What I've built so far proves that the concept works: AI can accurately diagnose plant diseases, and mobile technology can deliver that expertise directly to farmers who desperately need it. But a working prototype isn't a finished solution. To turn this into a tool that truly transforms lives, there's still a long road ahead and I'm ready to walk it.

Here's what comes next:

- 1. Expand the disease library.** Right now, the app covers a limited set of crops and conditions. To be truly useful, it needs to recognize a much wider range of plant diseases—especially those that hit local staples like cassava, maize, cocoa, and plantains. Every new disease we add is another harvest we help protect.
- 2. Train the AI with real-world data.** The models are only as good as the images they learn from. I plan to collect thousands of new photos directly from Cameroonian farms taken in different light, different seasons, and different stages of disease. That diversity will make the app smarter, more accurate, and more reliable in the messy, unpredictable conditions of real life.
- 3. Speak the farmer's language.** French and English aren't enough. For a farmer in the Northwest or Far North regions, help needs to come in *their* language Pidgin, Fulfulde, Ewondo, or Bamileke. Integrating local languages isn't just a feature; it's a bridge to true inclusion.
- 4. Work without the internet.** Connectivity is still a luxury in many villages. That's why offline functionality is non-negotiable. The app must be able to diagnose diseases and provide advice even when there's no signal because crops don't wait for Wi-Fi.
- 5. Build partnerships, not just software.** Technology alone doesn't change the world people do. That's why I'm committed to collaborating with agricultural cooperatives, NGOs, and government ministries. These partnerships won't just fund the next phase; they'll ensure that farmers actually hear about the app, trust it, and use it. Adoption is just as important as invention.

The prototype is proof of what's possible. The perspectives are proof of what's coming.

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IMPLEMENTATION OF A MOBILE APPLICATION FOR PLANT DIAGNOSIS



GENERAL CONCLUSION

This project started with a simple question: why should a farmer in rural Cameroon have to wait days for help when the answer could fit in their pocket? That question became the driving force behind everything you've just read. What I've designed here is not just a mobile app it's a response to a very real, very urgent problem. The application I've conceptualized and prototyped tackles one of the biggest bottlenecks in Cameroonian agriculture: the gap between farmers who need expert advice and the experts who can't reach everyone in time. By combining artificial intelligence with the growing accessibility of smartphones, this solution puts diagnostic power directly into the hands of those who need it most no long waits, no costly travel, no guesswork. Of course, this is still a prototype. There's more work ahead: refining the AI model, expanding the disease database, testing with real farmers in the field, and integrating local languages to make it truly inclusive. But the foundation is solid, and the vision is clear. For me personally, this project has been more than an academic exercise. It's been a chance to prove to myself and to others that technology can be a force for good when it's grounded in real human needs. I didn't need an internship to see that. I just needed to look at the farmers around me and ask what they were missing. I believe this application has real potential not just as a tool, but as a symbol of what's possible when innovation meets purpose. With continued development and the right support, I'm confident it can become a reference solution for agricultural support, first in Cameroon, and eventually beyond.

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